

Problems and artefacts

There are very many practical problems which arise when recording an EEG. Some of the problems are associated with the equipment, but other problems originate from the patient, whose co-operation must be obtained if spurious signals from movement of the eyes or muscles are to be avoided. The list which follows includes some of the more common causes of artefacts on an EEG trace: electrode artefacts are usually the most troublesome.

Eye potentials. There is a potential of several millivolts between the back and front of the eyes. This potential gives rise to current flow through the tissues surrounding the eyes and this current will change as the eyes move. The effect can be seen as large deflections of the EEG trace when the eyes are moving.

ECG. The ECG is not usually a major problem in EEG recording, but if the recording electrodes are spaced a long way apart an ECG will be recorded and seen as sharp regular deflections on the recording. The artefact which results if the patient has an implanted cardiac pacemaker is very large and cannot be removed.

Electrode artefacts. If the patient moves or the wires leading to the electrodes are disturbed, then the electrochemical equilibrium underneath the electrodes will be changed and so potential changes will occur. Another effect can occur if the patient is perspiring as this will also disturb the electrochemical equilibrium under the electrodes and give rise to quite large potential changes. These changes are usually slow baseline changes on the EEG.

There are very many more sources of spurious signals on the EEG trace. Ways in which electrical interference can arise were described in [Chapter 10](#) (section 10.4). It has even been suggested that problems have arisen from dental fillings, where an electrical discharge between different metallic fillings gave rise to artefacts in the EEG. The EEG technician must always be on guard for possible sources of interference.

Particular EEG patterns have been associated with many conditions such as cerebral tumours, epilepsy, haematomas, concussion and vascular lesions. However, no attempt will be made to describe these patterns here. Analysis of EEG signals is not easy because it is difficult to describe the signals. The ECG/EKG can be described in simple terms because there are only about five major components to the waveform, but the EEG is a much more complex signal. The various frequency ranges of the EEG have been arbitrarily assigned Greek letter designations to help describe waveforms. Electroencephalographers do not agree on the exact ranges, but most classify the frequency bands as follows: below 3 Hz, delta rhythm; from 3–7 Hz, theta rhythm; from 8–13 Hz, alpha rhythm; and from 14 Hz upwards, beta rhythm.

Most humans develop EEG patterns in the alpha range when they are relaxed with their eyes closed. The alpha rhythm seems to be the idling frequency of the brain and as soon as the person becomes alert or starts thinking the alpha rhythm disappears. This is the rhythm which is used in biofeedback systems where the subject learns to relax by controlling their own alpha rhythm.

Very many attempts have been made to analyse the EEG using computers, in order to help clinical interpretation. Many EEG machines include a frequency analyser which presents the frequency components of the EEG on the same chart as the EEG. Currently none of the methods of analysis have been found useful routinely and so they will not be described here.

Many EEG departments make EEG evoked response measurements in addition to the background EEG.

16.4. ELECTROMYOGRAPHIC (EMG) SIGNALS

An electromyograph is an instrument for recording the electrical activity of nerves and muscles. *Electro* refers to the electricity, *myo* means muscle and the graph means that the signal is written down. The electrical signals can be taken from the body either by placing needle electrodes in the muscle or by attaching surface electrodes over the muscle. Needle electrodes are used where the clinician wants to investigate neuromuscular disease

by looking at the shape of the electromyogram. He may also listen to the signals by playing them through a loudspeaker, as the ear can detect subtle differences between normal and abnormal EMG signals. Surface electrodes are only used where the overall activity of a muscle is to be recorded; they may be used for clinical or physiological research but are not used for diagnosing muscle disease. Both surface electrodes and needle electrodes only detect the potentials which arise from the circulating currents surrounding an active muscle fibre, and do not enable transmembrane potentials to be recorded. Nerves and muscles produce electrical activity when they are working voluntarily, but it is also possible to use an electrical stimulator to cause a muscle to contract and the electrical signal then produced is called an evoked potential. This is the basis for nerve conduction measurements, which allow the speed at which nerves conduct electrical impulses to be measured. This technique can be used to diagnose some neurological diseases and the principles of the method are explained in section 16.5.1.

Needle electrode measurements are almost always performed and interpreted by clinical neurologists, although both technical and scientific assistance may be required for the more sophisticated procedures. Nerve conduction measurements can be made as an unambiguous physiological measurement which is then interpreted either by medically or technically qualified staff.

16.4.1. Signal sizes and electrodes

The functional unit of a muscle is one motor unit but, as the muscle fibres which make up the unit may be spread through much of the cross-section of the muscle, it is impossible to record an EMG from just one unit. If a concentric needle electrode, of the type shown in figure 9.3, is placed in a weakly contracting muscle, then the EMG obtained will appear as in figure 16.21. Each of the large spike deflections is the summation of the muscle action potentials from the fibres of the motor unit which are closest to the tip of the needle electrode.



Figure 16.21. An EMG recorded via a concentric needle electrode in a weakly contracting striated muscle.

Remember that an upwards deflection represents a negative potential. The largest spikes all come from the same motor unit which is firing repetitively every 50 ms, but many other smaller spikes can be seen from fibres which are further away from the needle. The signal shown in figure 16.21 is a normal needle electrode EMG. The signal has a maximum amplitude of about $500\ \mu\text{V}$ and the frequency content extends from about 10 Hz to 5 kHz. If the strength of contraction of the muscle is increased, then more motor units fire and the spike repetition frequency is increased, but the frequency content of the signal will not change significantly.

If a more localized recording is required then a bipolar concentric needle electrode can be used (figure 16.22). With a monopolar concentric needle electrode the signal is recorded from between the tip of the needle and the shaft; with a bipolar needle, the signal is that which appears between the two exposed faces of platinum at the needle tip, and the shaft is used as the earth or reference electrode. A bipolar needle only records from the tissue within about 1 mm of the tip and the signals obtained are smaller and also have a higher-frequency content than concentric needle recordings.

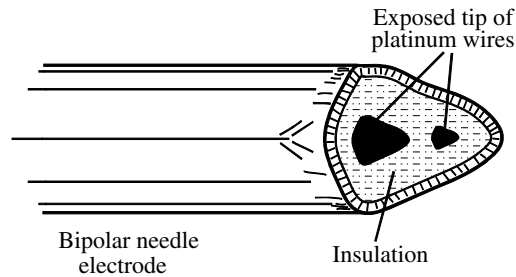


Figure 16.22. *The tip of a bipolar needle electrode.*

Any surface electrode placed over an active muscle can be used to record an EMG. If one electrode is placed over a muscle and the other electrode is placed several centimetres away, then EMG signals will be obtained from all the muscles lying between the electrodes. A more specific recording can be made if smaller electrodes are used and the separation is reduced to a few millimetres, but if the separation is reduced below about 4 mm then the amplitude of the signal falls rapidly. A very convenient and cheap electrode can be made from metal foil which can be cut to the desired size, will conform to the contours of the body and can be attached with adhesive tape. The skin must of course be cleaned and abraded before electrolyte jelly is applied and the electrode attached.

It is not possible, using surface electrodes, to record an EMG from just one muscle without interference from other muscles lying nearby. Even if two small electrodes are placed on the forearm, the EMG obtained will arise from many muscles. Localized recordings can only be made from needle electrodes, but these are uncomfortable and cannot be left in place for long periods of time. There is a type of electrode called a 'fine wire electrode' which can be left in a muscle for long periods: a wire is passed down the centre of a hypodermic needle which is then withdrawn, leaving the wire within the muscle. This can give an excellent long-term EMG recording.

The high-frequency content of surface electrode EMG signals is less than that from needle electrodes because of the volume conductor effects which were described in section 16.1.6. The recording amplifier should have a bandwidth from 10 to 1000 Hz. The amplitude of the signals depends upon the relative position of the electrodes and the muscle, but signals up to about 2 mV are typical.

16.4.2. EMG equipment

An EMG machine can be used to record both voluntary signals and evoked potentials. The amplitude of the signals will range from less than 1 μV up to 10 mV; the smallest signals are those produced by nerves and recorded from surface electrodes, whereas the largest are those evoked potentials from large muscles. [Figure 16.23](#) gives a block diagram of an EMG machine.

The pre-amplifier will be a differential amplifier with the following typical specification.

Amplification	100
Input impedance	10 M Ω
Noise with input shorted	2 μV p-p
Common-mode rejection ratio	80 dB
Bandwidth (-3 dB points)	10 Hz–10 kHz

The output from the pre-amplifier is taken to the main amplifier and then to the A–D converter and host computer. The signal is also usually taken to a loudspeaker as EMG signals fall within the audio band and the

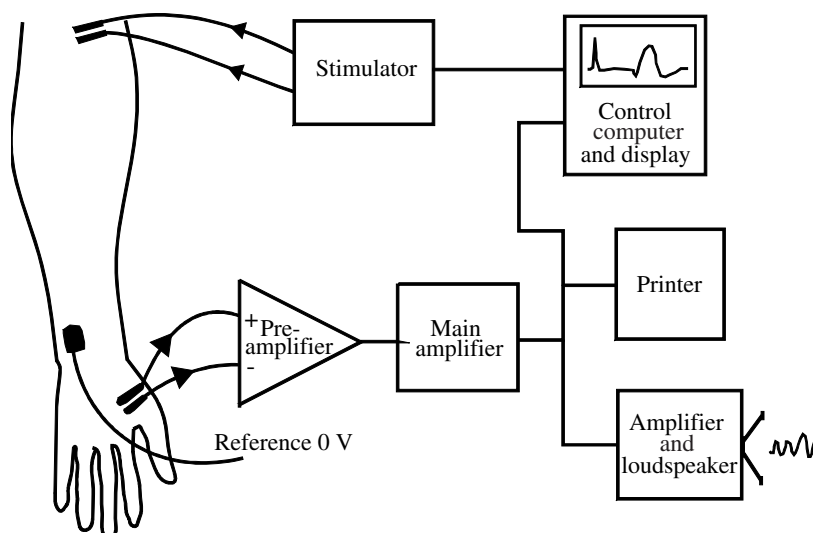


Figure 16.23. A block diagram of an electromyograph.

ear is very sensitive to subtle distinctions between signals. The remaining component of the EMG machine is the stimulator which is used for nerve conduction measurements. This will be considered in some more detail in section 16.5.

Equipment testing

The most common fault in all electrophysiological equipment is broken leads. Plugs and leads must be inspected regularly. Surface electrodes will have three connections to the differential amplifier: two inputs and an earth connection. A check on the operation of an EMG amplifier can be made by shorting together the three input connections and setting the amplifier gain to maximum. This may give a display of $10\ \mu\text{V}$ per division on the screen, and if the amplifier is operating correctly a band of noise will be seen on the trace. By increasing the volume control on the loudspeaker amplifier the noise should be heard as a random broad-band signal. The simplest way to check the stimulator is to hold both output connections in *one* hand and to increase the output control slowly. A shock should be felt at an output of about 60 V.

16.4.3. Normal and abnormal signals

Clinical electromyography using needle electrodes consists of inserting the sterilized needle into a muscle and then recording a voluntary EMG pattern from several points within the muscle. Samples are taken at several points because a diseased muscle may contain both normal and abnormal fibres. The neurologist will usually listen to the EMG signal, which sounds like intermittent gunfire. The patient will normally be asked to make only a mild contraction of the muscle so that individual spikes can be identified. When a strong contraction is made, a complete interference pattern is obtained sounding rather like an audience clapping. Not all muscles give the same sound although the difference between muscles of the same size is not great. An extreme case is the signals which can be recorded from a fine needle placed in the small muscles which move the eyes; these muscles are very small and the spikes obtained are of very short duration.

An individual action potential or spike when viewed on a screen has a total duration of a few milliseconds and usually contains only two or three deflections. In a myopathic muscle the action potentials are often smaller, may have more than three phases, and are of shorter duration than normal signals.

It is very difficult to distinguish individual spike potentials from a surface electrode recording. The amplitude of the EMG waveform is the instantaneous sum of all the action potentials generated at any given time. Because these action potentials occur in both positive and negative directions at a given pair of electrodes, they sometimes add and sometimes cancel. Thus the EMG pattern appears very much like a random noise waveform with the energy of the signal a function of the amount of muscle activity (figure 16.24).

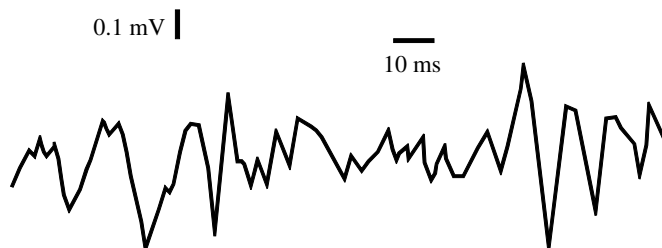


Figure 16.24. *An EMG recorded from surface electrodes.*

Signal analysis and clinical uses

Electromyography is used:

- in the diagnosis of neuromuscular disorders;
- as a measure of relaxation in the application of biofeedback techniques;
- as an index of muscle activity in physiological studies such as gait analysis.

There are very many clinical uses of electromyography, but it must be said that electromyography is really an extension of the classical methods of clinical examination, and each patient must be studied as an independent exercise in neurology. The skill of the electromyographer is as much in the planning of the examination as in its performance and interpretation. It seems improbable that electromyography will ever become a routine test performed by a technician under remote supervision.

Having made the point of the last paragraph it is of interest to outline very briefly the areas where clinical electromyography is useful. Following damage to a nerve, EMG signals give characteristic patterns of denervation which allow a prediction to be made about recovery. Damaged nerves may recover over periods as long as several years. EMG patterns characteristic of denervation include spontaneous activity such as small fibrillation potentials of short duration, instead of normal voluntarily produced spike potentials. Central neurogenic lesions such as motor neurone disease, poliomyelitis, and also spinal cord compression, cause characteristic EMG patterns which include large spike potentials with many deflections, synchronized motor unit activity, and some spontaneous electrical activity. Various inherited myopathies such as muscular dystrophy also give characteristic EMG patterns where the spike potentials are small, look ragged and contain more high-frequency components than the normal EMG.

Many methods of signal analysis have been tried to quantify EMG patterns; some depend upon measuring the frequency content of the signals. These can be of some use in quantifying the EMG and they have been shown to be helpful in identifying the carriers of muscular dystrophy, but they are not yet applied routinely.

Fatigue Characteristics of EMG

Introduction to EMG

- Electromyography (EMG) measures electrical activity of muscles.
- Used to assess muscle function and neuromuscular health.
- Surface EMG commonly used in biomechanics, rehabilitation, and ergonomics.

Muscle Fatigue

- Muscle fatigue is the decline in the ability of a muscle to generate force.
- Occurs during sustained or repetitive contractions.
- Caused by metabolic, neural, and mechanical factors.

Fatigue Characteristics of EMG

- Increase in EMG amplitude (RMS) during early fatigue.
- Decrease in mean and median frequency.
- Reduction in muscle fiber conduction velocity.
- Changes in motor unit recruitment and synchronization.

Amplitude Changes in EMG

- RMS value increases as additional motor units are recruited.
- Increased firing rate to maintain muscle force.
- In severe fatigue, amplitude may decrease due to inability to sustain contraction.

Frequency Domain Changes

- Power spectrum shifts toward lower frequencies.
- Median Frequency (MDF) decreases with fatigue.
- Mean Frequency (MNF) also decreases.
- Used as reliable indicators of localized muscle fatigue.

Physiological Causes

- Accumulation of lactic acid and metabolites.
- Decrease in pH and ionic imbalance.
- Reduced conduction velocity of muscle fibers.
- Neuromuscular transmission changes.

Methods to Analyze EMG Fatigue

- Time-domain analysis (RMS, Integrated EMG).
- Frequency-domain analysis (Median and Mean Frequency).
- Conduction velocity measurement.
- Spectral analysis using FFT.

Applications

- Sports science and performance monitoring.
- Rehabilitation and physiotherapy.
- Ergonomic assessment and workplace design.
- Prosthetics and biomedical research.

Conclusion

- EMG provides valuable indicators of muscle fatigue.
- Amplitude and frequency parameters are widely used.
- Useful in clinical, research, and engineering applications.

Muscle Stimulator

- Electrical device used to stimulate muscles using electrical impulses.
- Used in physiotherapy, rehabilitation, sports medicine, and biomedical engineering.
- Also called Electrical Muscle Stimulation (EMS).

Physiological Basis of Muscle Stimulation

- Muscle and nerve tissues are electrically excitable.
- Resting membrane potential exists across nerve and muscle membranes.
- External electrical stimulus can depolarize the membrane and trigger contraction.

Mechanism of Muscle Contraction

- Electrical stimulus depolarizes motor nerve.
- Signal reaches neuromuscular junction.
- Acetylcholine release triggers muscle fiber contraction.

Electrical Principles of Stimulation

- Effectiveness depends on amplitude, pulse duration, and frequency.
- Proper selection of parameters ensures safe and effective stimulation.

Strength–Duration Relationship

- Rheobase: Minimum current needed for excitation with long pulse duration.
- Chronaxie: Minimum pulse duration required at twice rheobase current.
- Denervated muscles require longer pulse duration and higher amplitude.

Types of Electrical Pulses

- Monophasic pulses: Current flows in one direction.
- Biphasic pulses: Alternating current direction, reduced skin irritation.
- Burst stimulation: Group of pulses used in strengthening.

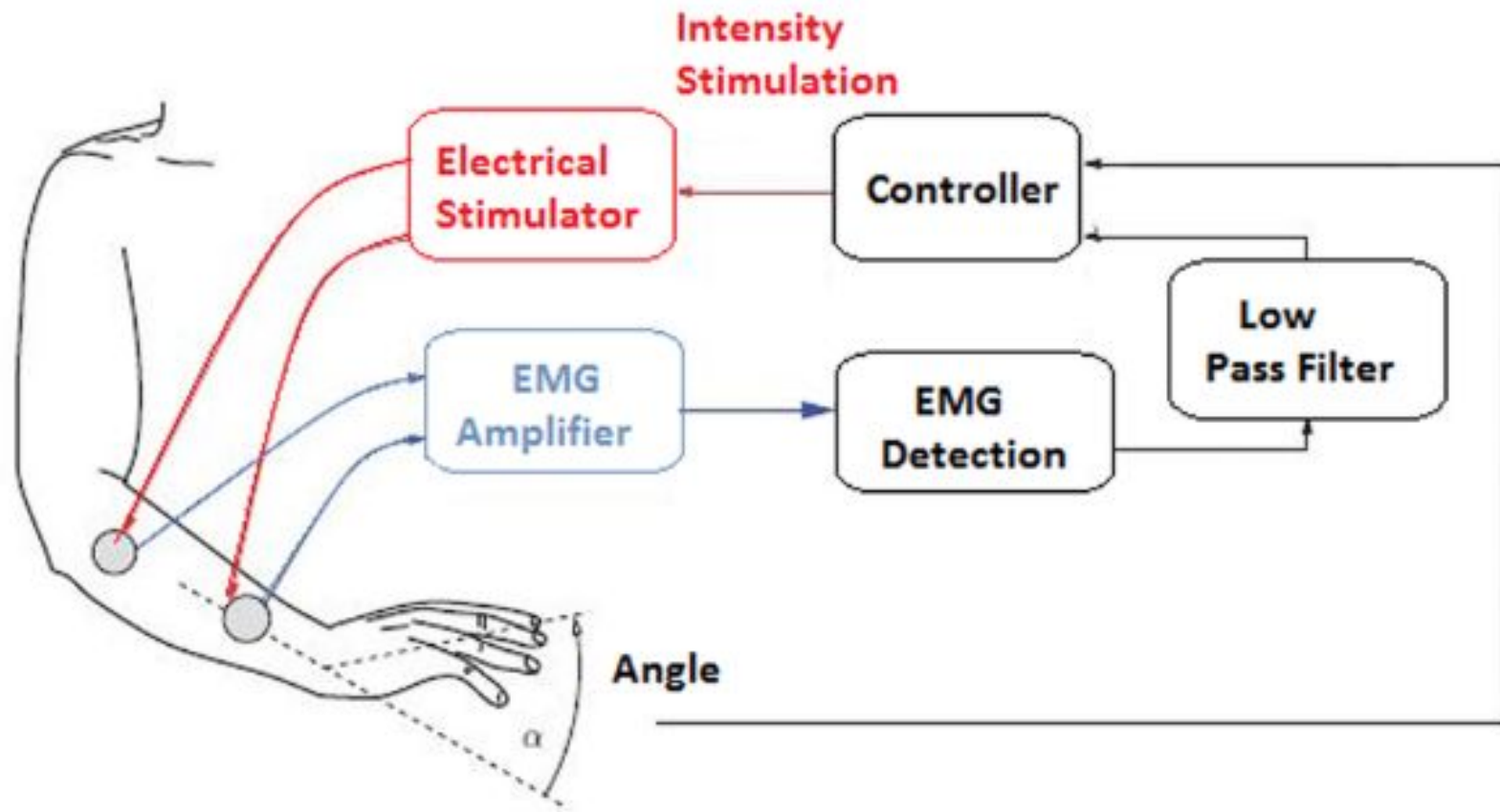
Stimulation Parameters

- Frequency: 1–150 Hz determines contraction type.
- Pulse width: 50–400 microseconds controls nerve excitation.
- Intensity: Controls strength of contraction.
- Duty cycle: Controls fatigue level.

Frequency Effects

- 1–10 Hz: Twitch contraction.
- 20–50 Hz: Smooth tetanic contraction.
- Above 80 Hz: Rapid fatigue may occur.

Block Diagram of Muscle Stimulator



Physiological Effects

- Improved blood circulation.
- Prevention of muscle atrophy.
- Improved neuromuscular coordination.
- Muscle strengthening over time.

Types of Muscle Stimulators

- EMS: Muscle strengthening and rehabilitation.
- TENS: Pain relief by stimulating sensory nerves.
- FES: Restoration of functional movement in paralysis.

Applications

- Stroke rehabilitation.
- Spinal cord injury therapy.
- Post-operative strengthening.
- Sports injury recovery.
- Pain management.

Advantages

- Non-invasive method.
- Portable and programmable devices available.
- Effective in rehabilitation and physiotherapy.

Limitations and Precautions

- Not recommended for patients with pacemakers without supervision.
- May cause skin irritation if improperly used.
- Proper electrode placement is essential for safety.

Conclusion

- Muscle stimulators are important electrotherapy devices.
- Proper parameter selection ensures safe and effective therapy.
- Widely used in clinical and biomedical applications.

Nerve Stimulator

Introduction

- A nerve stimulator is a biomedical device used to apply controlled electrical impulses to nerves.
- Used in anesthesia, physiotherapy, pain management, and diagnosis.
- Helps assess nerve and muscle function.

Principle of Operation

- Electrical stimulation causes depolarization of nerves.
- Action potential is generated.
- Motor nerve stimulation produces muscle contraction.
- Response helps evaluate nerve integrity.

Basic Components

- Pulse Generator – Produces electrical impulses.
- Electrodes – Deliver stimulation to nerves.
- Control Unit – Adjusts current, frequency, pulse duration.
- Power Supply – Battery or rechargeable source.

Types of Nerve Stimulators

- Peripheral nerve stimulator – Used in anesthesia.
- TENS – Used for pain relief and physiotherapy.
- Implantable stimulators – Used in chronic pain and neurological therapy.

Modes of Stimulation

1. Single Twitch Stimulation

- A single electrical pulse is applied and the muscle response is observed.

2. Train-of-Four (TOF) Stimulation

- Four pulses are delivered in rapid succession to assess neuromuscular blockade.

3. Tetanic Stimulation

- Rapid stimulation at high frequency to evaluate neuromuscular transmission.

4. Double Burst Stimulation

- Used to detect residual neuromuscular blockade more accurately.

Applications

- Monitoring neuromuscular blockade in anesthesia.
- Pain management using TENS.
- Muscle rehabilitation and physiotherapy.
- Diagnostic nerve conduction studies.

Advantages and Limitations

Advantages:

- Portable and easy to use
- Quick assessment of nerve function

Limitations:

- Incorrect electrode placement affects accuracy
- Possible skin irritation

Safety Precautions

- Check electrode placement before use.
- Avoid stimulation near the heart.
- Use appropriate current levels.
- Follow manufacturer guidelines.

Conclusion

- Nerve stimulators are important tools in diagnosis, therapy, and monitoring.
- Widely used in anesthesia, physiotherapy, and pain management.

16.5. NEURAL STIMULATION

Because the body is a good conductor of electricity, and because our nerves and muscles function electrically, we would expect to see physiological effects when current is applied to the body. These effects can be a source of hazard but they can also be utilized both for the diagnosis and treatment of disease.

Chapter 8 dealt with the physiological effects of electricity. The three most important physiological effects of electricity were shown to be electrolysis, neural stimulation and heating. Neural stimulation was dealt with in some detail in section 10.2.1, but we will consider two applications of neural stimulation in this section.

16.5.1. Nerve conduction measurement

An evoked response can be recorded from a nerve following electrical stimulation. The response has the form shown in figure 16.12 and, by measuring the time between the stimulus and the response, a conduction time and velocity can be determined. The average nerve conduction velocity is about 50 m s^{-1} or 112.5 mph. This velocity is actually very slow and certainly far slower than the speed at which electrical signals travel down a length of wire; this speed approaches the speed of light. Just how slow a nerve conduction velocity is may be appreciated by thinking about what happens to the nerve impulses which would arise from your feet, if you jumped off a very high building. If the building were 200 m high then you would hit the ground at about 63 m s^{-1} (we have neglected air resistance in making this calculation) which is a higher velocity than that at which the nerve signals would travel towards your brain, so that you would not feel your feet hit the ground. However, the slow conduction velocity of nerves does not appear to handicap the human body and it does make measurement of the velocity very easy. 50 m s^{-1} is the conduction velocity of myelinated nerve fibres, but there are also non-myelinated fibres which transmit signals much more slowly. At birth, myelination is not complete and this is one of the reasons why nerve conduction velocities increase over the first few years of life. Because the velocity is changing rapidly before birth a premature baby will have a slower velocity than a full term child; indeed, it is possible to determine the gestational age of a child at birth by measuring the nerve conduction velocity. In figure 16.25 the conduction velocity of the posterior tibial nerve is shown for groups of children up to the age of one year.

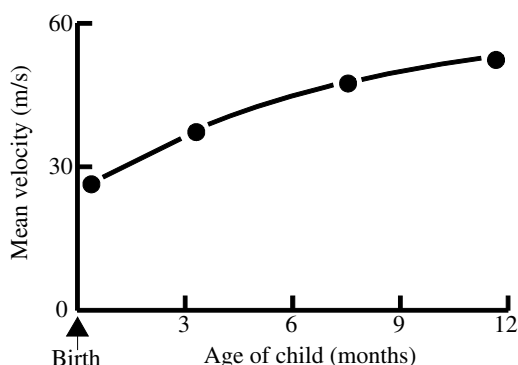


Figure 16.25. Nerve conduction velocity in the posterior tibial nerve is shown as a function of age. The points shown are mean values for a group of normal children. From B H Brown, R W Porter and G E Whittaker (1967) *Nerve conduction measurements in Spina Bifida Cystica*, 11th Annual Report of the Society for Research into Hydrocephalus (Carshalton).

Measurement of motor nerve conduction velocity

Following an electrical stimulus to a nerve, an action potential will travel down the nerve, along the terminal nerve fibres, across the neuromuscular junction and then cause a muscle action potential to spread along the muscle fibres. All these processes take some time and yet only the total time can be determined. This total time is called the 'latency'. However, by stimulating the nerve at two points and recording the difference in latency between the two we can determine the conduction time between the points of stimulation. Figure 16.26 shows where electrodes may be placed and equipment attached to make a measurement from the ulnar nerve, and figure 16.27 shows the results that should be obtained.

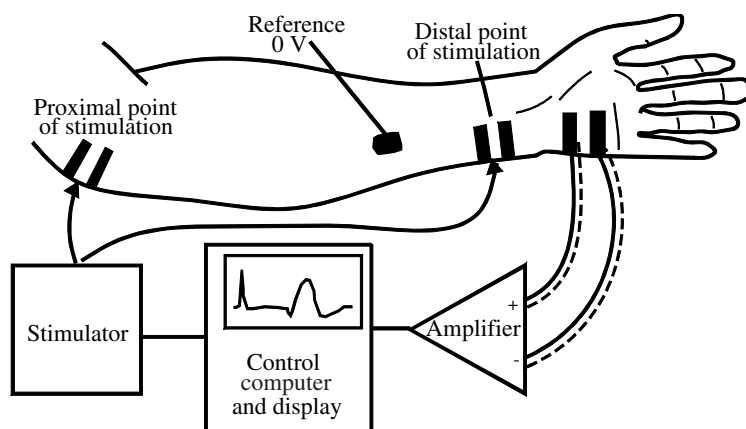


Figure 16.26. The system used to record an ulnar nerve conduction velocity.

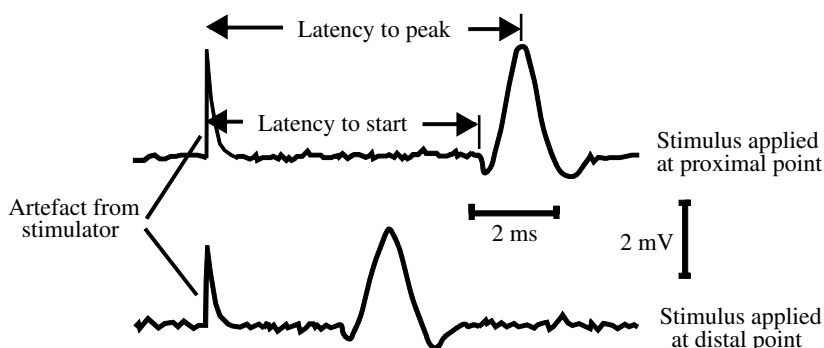


Figure 16.27. Recordings made following stimulation of a nerve trunk at two points.

Proximal means the point closest to the point of attachment of the limb to the body, and distal refers to the point which is further away; distance is measured between the stimulation cathodes because the action potential is initiated underneath the cathode. Latency is measured to the start of the muscle action potential by most workers, but to the major negative peak (upwards deflection) by some workers. There are established routines in every neurology department and 'normal' values will have been obtained for the particular method of recording.

One particular regime is now listed for recording a median nerve motor conduction velocity. The median nerve is one of the three major nerves which supply the arm; the other two are the ulnar and the radial. Figure 16.28 shows the major nerve trunks of the upper and lower limbs.

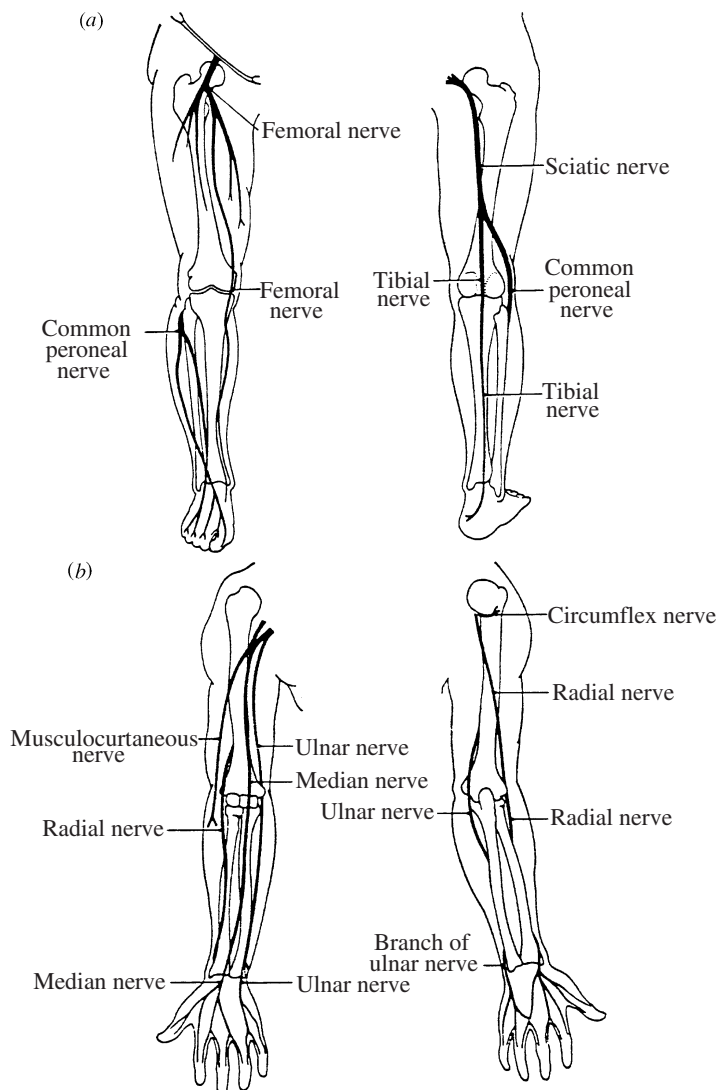


Figure 16.28. (a) *The major nerve trunks of the legs;* (b) *the major nerve trunks of the arms.*

Make sure that the patient is warm. Nerve conduction velocity decreases by several per cent for each centigrade degree drop in body temperature.

Prepare the skin by abrasion over the antecubital fossa, the wrist and the thenar muscles.

Make up three pairs of metal foil electrodes on adhesive tape, apply a small amount of electrode jelly to these and then attach them to the skin.

Set the EMG machine amplifier to 2 mV per division, the filters to a bandwidth of at least 10 Hz to 5 kHz and the time base to 2 ms per division. Check that the stimulator output is set to zero.

Connect the distal pair of electrodes to the stimulator with the cathode closest to the hand. Connect the amplifier to the electrodes on the hand, with the non-inverting input to the most distal electrodes; this will give a positive deflection for a negative signal underneath the first recording electrode. The leads to the electrodes should be secured to the arm, so that movement of the arm does not cause electrode movement artefact.

Prepare a skin site on the forearm and attach the earth or reference electrode. The patient may be lying down with the arm resting on the bed. Alternatively, they can sit on a chair and rest their arm on a convenient support about 20 cm above the level of the chair. It is important that the patient be asked to relax, because tension and the resultant continuous EMG activity make measurement difficult.

Obtain a continuous trace on the screen and check that no 50 Hz interference is present. If interference is present, first check that the electrodes have been applied well before investigating the many other possible causes of the interference.

Set the stimulator to give 100 μ s width pulses at 1 pps. Increase the stimulus amplitude progressively until the EMG response shows no further increase in amplitude. You then have a supramaximal response.

Measure the latency of the response.

Turn the stimulator output down to zero.

Now apply the stimulus to the upper pair of electrodes and repeat the previous three steps to obtain a value for the distal latency.

Disconnect the stimulator and amplifier and then record the distance between the two stimulation cathodes.

Calculate the conduction velocity.

Measurement of sensory nerve conduction velocity

The EMG signals obtained when a motor nerve is stimulated are several millivolts in amplitude and therefore relatively easy to record. When a sensory nerve is stimulated, the only signal obtained is that which arises from the circulating currents surrounding the active nerve, and this signal is very small. The signals are nearly always less than 10 μ V and a signal averager is required to reduce the background noise (see section 10.3.4).

There is one situation where a relatively large sensory nerve action potential is obtained and this can be useful for demonstration and teaching purposes. Following stimulation of the median nerve at the wrist, nerve impulses will be conducted in both directions along the nerve fibres; conduction up the sensory nerves in the normal direction is termed orthodromic conduction and conduction in the opposite direction is called antidromic conduction. Antidromic conduction gives rise to quite a large nerve action potential if recording electrodes are placed around the base of the index finger and a stimulus applied to the median nerve at the wrist. This signal can be recorded without a signal averager and should have a latency of about 5 ms to the peak of the action potential.

Accurate measurement of sensory nerve conduction velocities is often difficult because two convenient sites for stimulation of the same nerve are not available. For this reason only the latency may be measured and this value compared with a 'normal' range of values; alternatively an arbitrary reference point may be found on the nerve action potential waveform and the latency to this point used to calculate a 'velocity', using only a single nerve action potential recording.

The equipment used for signal averaging has already been described in the sections on evoked response measurement in [Chapter 10](#). No further details will be given here.

Possible measurements and clinical value

If great care is taken, then both sensory and motor conduction velocity measurements can be made from most of the major nerves in the body. Needle electrodes are needed in order to obtain sufficiently large signals in some cases. The following short list is of the stretches of nerve from which recordings can be made easily using only surface electrodes.

Motor measurements. Arm, median and ulnar nerves—axilla to elbow to wrist to hand; Leg, posterior tibial—behind the knee to the ankle to the foot; anterior tibial - head of the fibula to the ankle to the foot.

Sensory measurements. Arm, median nerve index finger to wrist; ulnar nerve—little finger to wrist.

When used in conjunction with EMG recording, nerve conduction measurement is an extremely useful diagnostic tool. The major use of nerve conduction measurements in isolation is in the diagnosis and investigation of entrapment neuropathies. The most common of these is ‘carpal tunnel syndrome’ where compression of the median nerve at the wrist causes an increased latency for both motor and sensory signals. Compression within the tarsal tunnel at the ankle can also be investigated by making MCV measurements. A further use of these measurements is in cases of nerve lesions, e.g. damage to the ulnar nerve at the elbow; stimulation above and below the site of the suspected lesion may show slowing across the lesion.

One final point should be made, that nerve conduction velocity is actually the conduction velocity of many nerve fibres not all of which will have the same conduction velocity. There is a range of conduction velocities and there are methods of measuring this spectrum, but these methods are not yet widely used.

16.6. PROBLEMS

16.6.1. Short questions

- a Which ion is actively transported out of a resting cell?
- b What is the normal resting potential across a human nerve cell: -153 , -70 or -15 mV?
- c What type of relationship is there between the intensity of sensation and pulse repetition rate in the human nervous system?
- d The largest component of the ECG/EKG is the R wave. To what mechanical activity of the heart does the R wave correspond?
- e There is a delay in the electrical conduction system of the heart between the upper and lower chambers (the atria and ventricles). What is the function of this delay?
- f What is the name of the insulating material that surrounds many human nerves?
- g What are nerve membranes made from?
- h What is meant by saying that smooth muscle is intrinsically active?
- i Will a monopolar recording of the potential produced in a volume conductor by a single wavefront of depolarization have a biphasic or triphasic shape?
- j What advantage does myelination of a nerve axon confer on a nerve cell?
- k What amplitude and frequency range would you expect for electrical signals generated within the body?
- l How long would a typical nerve action potential take to travel from the brain to the feet?
- m What is the approximate amplitude of the ECG/EKG when recorded from the skin?
- n What event does the T wave of the ECG/EKG correspond to?
- o What is the physiological function of acetylcholine?
- p Are the fastest conducting myelinated nerve fibres larger or smaller than the slowest fibres?

GAIT AND LIMB POSITION

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GAIT ANALYSIS

- Gait analysis - Functional evaluation of walking or running style
- Gait evaluation identifies - Functional limitations and Chronic pain related to physical activity

PHASES OF GAIT CYCLE

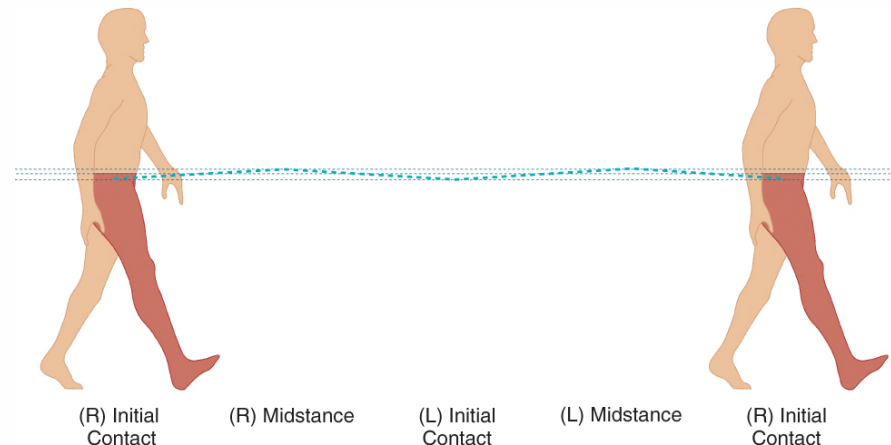
- Two distinct phases occur
 - Weight-bearing (WB) stance phase
 - Non–weight-bearing (NWB) swing phase
- Legs alternate between supportive (stance) and nonsupportive (swing).

PHASES OF GAIT CYCLE

Stance Phase (Weight Bearing)					Swing Phase (Non-weight Bearing)		
							
Initial Contact	Loading Response	Midstance	Terminal Stance	Preswing	Initial Swing	Midswing	Terminal Swing

PHASES OF GAIT CYCLE

- Efficient gait
 - Minimal side-to-side motion
 - Maximal forward motion
 - Body rises and falls approx. 5 cm
- Center of gravity
 - Path is a sinusoidal curve.



STANCE PHASE

- Five periods
 - Initial contact
 - Loading response
 - Midstance
 - Terminal stance
 - Preswing

SWING PHASE

- Three periods
 - Initial swing
 - Midswing
 - Terminal swing

EMG Biofeedback Instrumentation

- Electromyography (EMG) biofeedback is a technique used to monitor and provide feedback of muscle activity.
- It helps patients gain voluntary control over muscle function.
- Widely used in rehabilitation, physiotherapy, and neuromuscular training.

Introduction to EMG Biofeedback

- EMG measures electrical activity generated by skeletal muscles.
- Biofeedback converts this electrical activity into visual or auditory signals.
- Patients learn to modify muscle activity using real-time feedback.

Principle of Operation

- Muscle contraction generates electrical potentials.
- Surface electrodes detect these signals from the skin.
- Signals are amplified, filtered, processed, and displayed as feedback.

Block Diagram of EMG Biofeedback System

- Surface electrodes acquire EMG signals.
- Preamplifier increases low-level signals.
- Filters remove noise and interference.
- Signal processing converts EMG into usable feedback.
- Display or audio system provides feedback to the patient.

Electrodes Used in EMG

- Surface electrodes are commonly used for biofeedback applications.
- Silver/Silver-Chloride electrodes provide stable recordings.
- Proper skin preparation reduces impedance and improves signal quality.

Preamplifier and Instrumentation Amplifier

- EMG signals are very small (microvolt range).
- Instrumentation amplifiers provide high gain and high common-mode rejection ratio (CMRR).
- Isolation is used to ensure patient safety.

Filters in EMG Instrumentation

- High-pass filters remove motion artifacts.
- Low-pass filters reduce high-frequency noise.
- Notch filters eliminate power-line interference (50/60 Hz).

Signal Processing in EMG Biofeedback

- Rectification converts bipolar EMG signals to unidirectional signals.
- Integration or smoothing produces an envelope of muscle activity.
- Processed signal is easier to interpret and display.

Feedback and Display Systems

- Visual feedback through meters, graphs, or computer displays.
- Auditory feedback using tones that vary with signal strength.
- Helps patients understand and control muscle contractions.

Applications of EMG Biofeedback

- Rehabilitation after stroke or nerve injury.
- Treatment of muscle tension and pain disorders.
- Training in sports medicine and physiotherapy.
- Post-operative muscle re-education.

Advantages of EMG Biofeedback

- Non-invasive and safe technique.
- Provides real-time information on muscle activity.
- Enhances patient participation in therapy.

Limitations of EMG Biofeedback

- Susceptible to motion artifacts and noise.
- Requires proper electrode placement and calibration.
- Interpretation may require trained personnel.

Safety Considerations

- Electrical isolation is essential to protect patients.
- Proper grounding and shielding reduce interference.
- Regular calibration and maintenance ensure reliable operation.

Instruments for Surgery

■■■■► 27.1 PRINCIPLE OF SURGICAL DIATHERMY

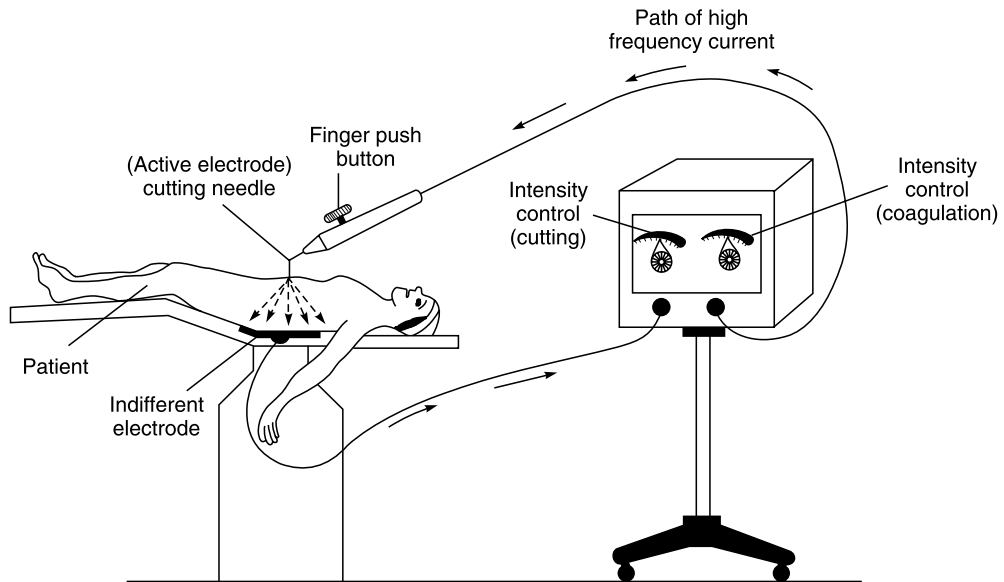
High frequency currents, apart from their usefulness for therapeutic applications, can also be used in operating rooms for surgical purposes involving cutting and coagulation. The frequency of currents used in surgical diathermy units is in the range of 1–3 MHz in contrast with much higher frequencies employed in short-wave therapeutic diathermy machines. This frequency is quite high in comparison with that of the 50 Hz mains supply. This is necessary to avoid the intense muscle activity and the electrocution hazards which occur if lower frequencies are employed. The power levels required for electrosurgery are below the threshold of neural stimulation provided that the diathermy frequency is in the radio-frequency range. When the frequency is at least 300 kHz, both the faradic and the electrolytic effects are largely eliminated during the flow of current through the human tissue. This then allows the exclusive utilization of the thermal effect in high frequency surgery providing both the applications for cutting and coagulation.

For their action, surgical diathermy machines depend on the heating effect of electric current. When high frequency current flows through the sharp edge of a wire loop or band loop or the point of a needle into the tissue (Fig. 27.1), there is a high concentration of current at this point. The tissue is heated to such an extent that the cells which are immediately under the electrode, are torn apart by the boiling of the cell fluid. The indifferent electrode establishes a large area contact with the patient and the RF current is therefore, dispersed so that very little heat is developed at this electrode. This type of tissue separation forms the basis of electrosurgical *cutting*.

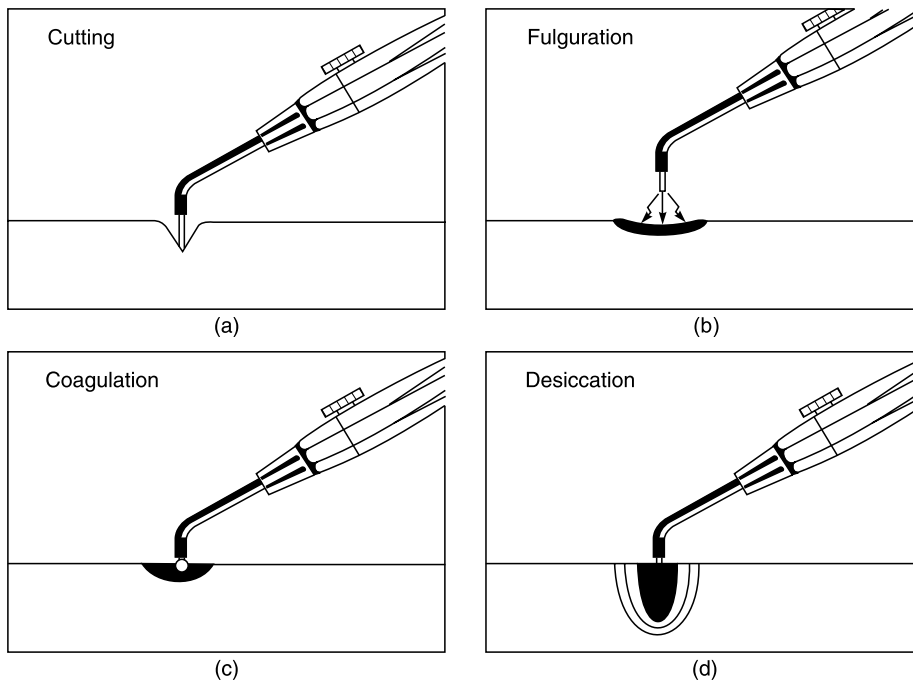
Electrosurgical *coagulation* of tissue is caused by the high frequency current flowing through the tissue and heating it locally so that it coagulates from inside. The coagulation process is accompanied by a grayish-white discoloration of the tissue at the edge of the electrode.

The term '*fulguration*' refers to a superficial tissue destruction without affecting deep-seated tissues. This is undertaken by passing sparks from a needle or a ball electrode of small diameter to the tissue. When the electrode is held near the tissue without touching it, an electric arc is produced, whose heat dries out the tissue. Fulguration permits fistulas and residual cysts to be cauterized and minor haemorrhages to be stopped.

In *desiccation*, needle-point electrodes are stuck into the tissue and then kept steady. Depending upon the intensity and duration of the current, a high local increase in heat will be obtained. The tissue changes due to drying and limited coagulation. Fig. 27.2 shows various types of electrosurgery techniques that are commonly employed in practice.



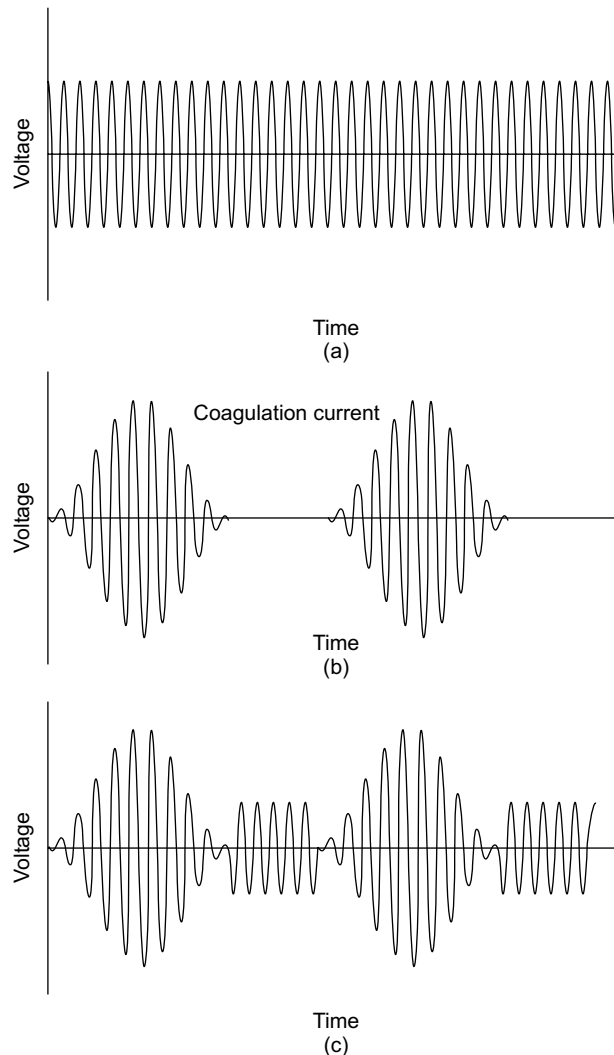
► Fig. 27.1 Principle of surgical diathermy machine



► Fig. 27.2 Various types of electro-surgery techniques commonly employed in practice

The concurrent use of continuous radio-frequency current for cutting and a burst wave radio-frequency for coagulation is called *Haemostasis* mode.

Different types of waveforms have been used to produce different effects for surgical procedures. Fig. 27.3 shows typical waveforms found in commercial equipment for cutting and coagulation. The cutting current usually results in bleeding at the site of incision, whereas the surgeon would require bloodless cutting. The machines achieve this by combining the two waveforms shown in Fig. 27.3 (c). The frequency of this blended waveform is generally the same as that used for cutting current.



► **Fig. 27.3** Types of waveforms generated by surgical diathermy machines: (a) Cutting current: a pure sine wave (b) Coagulate current: intermittent bursts of high voltage current (c) Blended current: A combination of cutting and coagulation currents.

The use of high frequency current offers a number of important advantages. The separation of tissues by electric current always takes place immediately in front of the cutting edge and is not caused by it. Electric cutting, therefore, does not require any application of force. Instead it facilitates elegant and effortless surgery. The electrode virtually melts through the tissue instantaneously and seals capillary and other vessels, thus preventing contamination by bacteria. A simplified method of coagulation saves valuable time since bleeding can be arrested immediately by touching the spot briefly with the coagulating electrode. A high frequency apparatus is regarded as standard equipment in the operating theatre and it is kept ready for all surgical interventions even if high frequency surgery is not intended so that use can be made of its superior method of electro-coagulation.

Biological tissue can only be cut when the voltage between the cutting electrode and the tissue to be cut is high enough to produce electric arcs between the cutting electrode and the tissue. The temperatures produced at the points at which the electric arcs contact the tissue like microscopic flashes of lightning are so high that the tissue is immediately evaporated or burned away. A voltage of approximately $200 V_p$ is required in order to produce the electric arc between a metal electrode and biological tissue. If the voltage is less than $200 V_p$, the electric arcs cannot be triggered and the tissue cannot be cut. The voltage suitable for cutting biological tissue ranges between $200 V_p$ and $500 V_p$. If the voltage rises above $500 V_p$, the electric arcs become so intense that the tissue is increasingly carbonized and the cutting electrode may be damaged. A visible arc forms when the electric field strength exceeds 1 kV/mm in the gap and disappears when the field strength drops below a certain threshold level.

Biological tissues are coagulated by thermal means if the requisite temperature is maintained at around 70°C .

■■■■► 27.2 SURGICAL DIATHERMY MACHINE

Basically, a surgical diathermy machine consists of a high frequency power oscillator. The earlier types of diathermy machines consisted of spark-gap oscillators whereas the current practice is to use thermionic valves or solid-state oscillators. A majority of the earlier units have access to both these power sources, viz. an RF generator and a spark-gap generator. The RF generator provides an undamped high frequency current (typically 1.75 MHz) which is suitable for making clean cuttings. The spark-gap generator produces damped high frequency current which is specifically suitable for the coagulation of all kinds of tissues. The mixing of both these currents signifies one of the most important possibilities for use in electrosurgery. By blending the currents of the tube and spark-gap generator, the degree of coagulation of wound edges may be chosen according to the requirements.

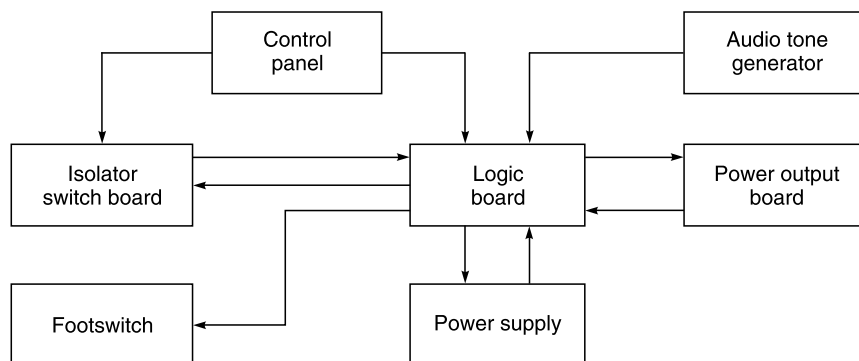
Whilst the detailed waveform and frequency spectrum used varies from one manufacturer to the other, the basic concept requires a high temperature arc, (Dobbie, 1969) at the operative site. In practice, the cross-section of the arc is extremely small, considerably less than 1 mm diameter, leading to a high current density in the arc. As the heating effect is proportional to the square of the current density, the effect is localized to form the arc. Other factors affecting the rise in temperature are the composition of the tissues and the magnitude of cooling provided by the local blood flow or any other heat transport system.

Despite the fact that surgical diathermy machines have been routinely employed for over half a century, the most significant technological developments have occurred within the past decade only. Solid-state generators have replaced a substantial number of vacuum tube

and spark-gap units. Disposable, self-adhering dispersive electrodes (generally known as 'ground pads') are now widely used in place of the large area buttplate. A number of design features have enhanced safety. These features include a variety of circuit integrity monitors like dispersive electrode cable continuity, patient circuit continuity and alternate-path current monitors, among others.

The frequency of operation of solid-state diathermy machines is 250 kHz to 1 MHz. They deliver 400 W in 500 Ω load at 2000 V in the cutting mode and around 150 W in the coagulation mode. In coagulation, the burst duration is 10–15 s and repetition frequency of the burst is 15 kHz.

Fig. 27.4 shows a block diagram of the solid-state surgical diathermy machine. The heart of the system is the logic and control part which produces the basic signal and provides various timing signals for the cutting, coagulation and haemostasis modes of operation. An astable multivibrator generates 500 kHz square pulses. The output from this oscillator is divided into a number of frequencies by using binary counters. These are the frequencies which are used as system timing signals. A 250 kHz signal provides a split phase signal to drive output stages on the power output circuit. A 15 kHz gating signal produces the repetition rate for the three cycles of the 250 kHz signal which make up the coagulating output. The pulse width of this output is set at about 12 μ s.



► Fig. 27.4 Block diagram of solid state electro-surgical unit

The 250 kHz signal used for cutting is given to power output stage where it controls the push-pull parallel power transistor output stage. The output of this high power push-pull amplifier is applied to a transformer which provides voltage step-up and isolation for the output signal of the machine. In order to meet the high power requirements, as much as 20 transistors are used in a parallel Darlington circuit. However, the power output amplifier circuitry varies considerably among the different commercial equipment. The modern machines employ both bi-polar junction transistors and power metal oxide-semiconductor field-effect transistors (MOSFET) in a cascade configuration or the use of a bridge connection of MOSFETs.

In order to facilitate identification of each mode of operation, the machines incorporate an audio tone generator. The tone signals are derived from the counter at 1 kHz (coagulation), 500 Hz (cutting) and 250 Hz (haemostasis). The isolator switch provides isolated switching control between the active hand switch and the rest of the unit. A high frequency transformer coupled power oscillator is used in which isolated output winding produces a DC voltage. The load put on the DC output by the hand switch is reflected back to the oscillator, accomplishing isolated switching. There is a provision to interrupt the power output if so desired.

Besides these basic functional circuits, logic circuits are used to receive external control signals and to operate the isolating relays, give visual indications and determine the alarm conditions. The logic circuits receive information from the foot-switch, finger switch and alarm sensing points. A thermostat is sometimes mounted on the power amplifier heat sink. In case of over temperature, it becomes open-circuited, signalling an alarm and interrupting the output.

The output circuit in the diathermy machine is generally isolated and carefully insulated from low frequency primary and secondary voltages. Blocking capacitors serve to effectively prevent any low frequency from appearing in the output circuit, and the isolated output reduces the possibility of burns due to an alternate path to ground. Complaints of electrical shock during surgery can almost always be attributed to muscle contractions of the patient. This is caused by the rectification of the high frequency energy at the junction of the active electrode and the tissue in the presence of an arc, which is the actual means of performing electrosurgery. This phenomenon is observed most when operating in a site of sensitive nerve tissue. There is, however, no danger to the patient or to the operator due to this action. On the other hand, anyone in close proximity to the radio-frequency carrying cables or electrodes will have some energy induced into his body. If by chance, he touches the metal cabinet of the surgical unit or any other conductive surface, current will flow through his body, resulting in a spark at the point of contact. It is advisable to avoid contacts with conducting surfaces by those who happen to be near the machine or cables.

The gases used in anaesthesia tend to settle near the floor. Therefore, the construction of the foot switch should be such that no explosion should occur in the atmosphere surrounding this switch caused by the operation of the electrical contacts within the switch.

Solid-state machines mostly incorporate an independent bi-polar RF generator for microsurgery procedures offering a fine output power control. The output waveform is a damped sinusoid at a repetition frequency of 30 kHz. The power output across 300 Ω is about 20 W, with peak to peak voltage of 1500 V open circuit.

27.2.1 Automated Electrosurgical Systems

Haag and Cuschieri (1993) illustrated (Fig. 27.5) that with a conventional electrosurgical unit, there is a considerable fluctuation of the output voltage throughout the 3-s period of the cut. The cause of this undesirable fluctuation is linked to the following factors:

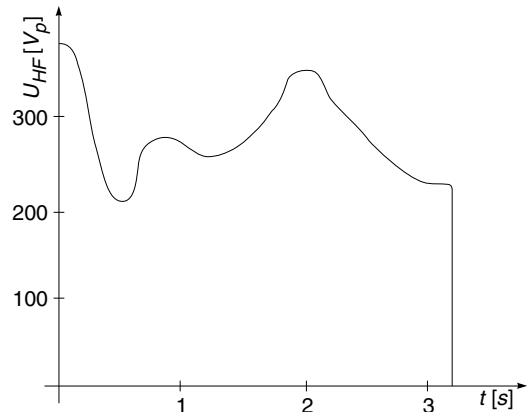
- *Size and Shape of the Cutting Electrode:* The conditions are different for the generator if, for instance, cutting is performed with electrode of large surface area or with a fine needle.
- *Type and Speed of Cut:* The cutting quality is determined by the speed with which the electrode is moved (quick or slow) and by the type of cut (superficial or deep)
- *Different Tissue Properties:* The tissue itself has a strong influence on the quality of the cut. For example, in tissues with a high resistance such as fat, the output voltage is increased whereas in tissues with a low electric resistance, such as nerves and blood vessels, the output voltage may drop significantly.

The variations in the output voltage due to the above factors considerably affect the quality of the cut. At times, the maximum output voltage can become so high (above 600°C) that severe carbonization occurs. Conversely, the minimum value of the output voltage can become so low (below 200°C) that cutting action is not achieved. In order to overcome this problem, microprocessor- controlled automated systems have been developed so that the output voltage or the spark intensity remains constant.

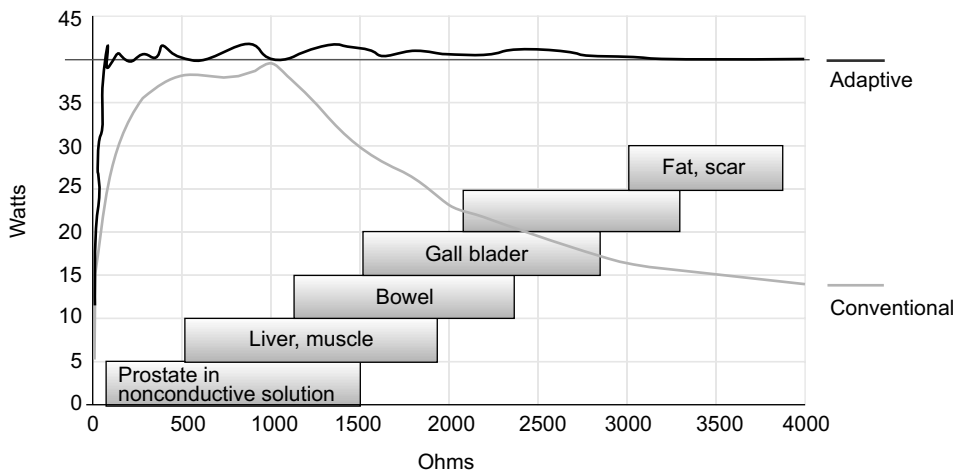
In this machine, the variables—current, tissue resistance, voltage and spark intensity—are registered by means of an inbuilt sensor system and then processed as defined output signals. The automatic control operates on two different criteria:

- *Voltage control*: whereby the selected voltage is controlled and held constant.
- *Spark control*: by which the selected spark intensity is held constant.

Today's electrosurgical generators use closed-loop control loops to adjust the voltage and current to keep the output power constant as the active monopolar electrode moves through tissues of varying impedance. These "adaptive" generators (Massarweh *et al.*, 2006) are a significant improvement over those used in traditional electrosurgery (Fig. 27.6).



► **Fig. 27.5** Cut made with a conventional electrosurgical unit. The incision is uneven with variable coagulation depth along its length due to considerable fluctuation of the voltage secondary to the source resistance of the machine



► **Fig. 27.6** The difference between an "adaptive" generator and one that does not sense the impedance and adjust accordingly (Adapted from Massarweh *et al.*, 2006)

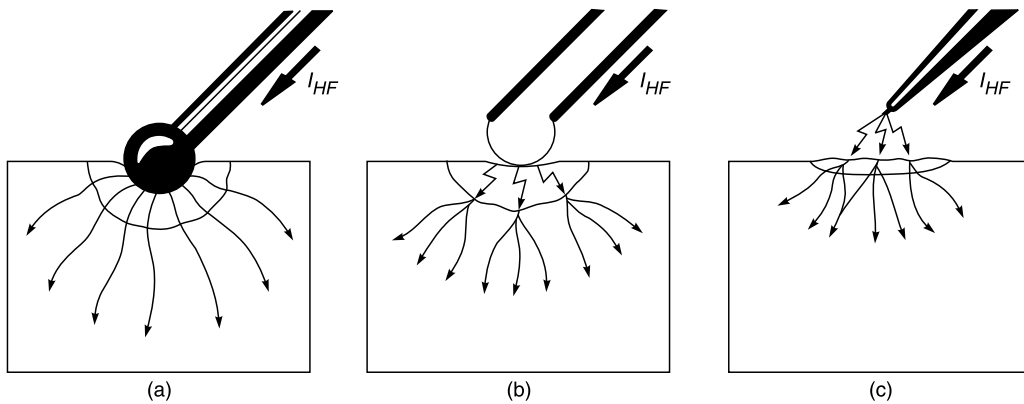
The control of spark intensity is relatively complex because of its non-linear nature. It has been revealed that a number of parameters are directly proportional to the spark intensity and these are used to pre-select and maintain constant these non-linear variables. The design of the control system ensures that the cutting quality is independent of size and shape of the electrode, the type and speed of the cut and the varying tissue properties.

Apart from ensuring a good quality of the cut, the microprocessor-controlled machine also provides the following coagulation modes:

- **Soft coagulation [Fig. 27.7(a)]**: In this, no electric arcs are produced between the coagulation electrode and the tissue during the entire coagulation process to prevent the tissue

from becoming carbonized. Soft coagulation is recommended in which coagulation electrodes are brought in direct contact with the tissue to be coagulated.

- **Forced Coagulation [Fig. 27.7(b)]:** This is characterized by the fact that electric arcs are intentionally generated between the coagulation electrode and the tissue in order to obtain deeper coagulation than could be achieved with soft coagulation, particularly when using thinner or smaller electrodes.
- **Spray Coagulation [Fig. 27.7(c)]:** In this, electric arcs are deliberately produced between the spray electrode and tissue so that direct contact between electrode and tissue becomes unnecessary. Spray coagulation is used both for surface coagulation and haemostasis of vessels not directly accessible to coagulation electrodes, such as those hidden in bone fissures.



► **Fig. 27.7** (a) Soft coagulation (b) forced coagulation (c) spray coagulation

The microprocessor-controlled machines provide numerous safety features such as an error detection system, an error signalling system and an error storage system. If a certain error arises, it is immediately displayed and the cause of the error can be readily determined. A variety of safety features which help in reducing the risks in high frequency surgery for both the patient and the users are provided in the machines. These features include low frequency leakage current monitor, high frequency leakage current monitor, output error monitoring, time limit monitoring, operating errors and neutral electrode safety system.

Modern surgical diathermy machines are programmable and user-friendly. For instance, frequently used standard settings can already be programmed by the manufacturer before delivery and individual customized settings can easily and swiftly be programmed later. Some machines have a power peak system that delivers a very short power peak at the beginning of electro-surgical cutting to start the cutting arc. Thereafter, average power can be limited to relatively small amounts, which signifies an improvement in protection against unintentional thermal tissue damage. Continuous monitoring of current and voltage levels and making automatic adjustment under the control of a microprocessor provides for a smooth cutting action throughout the procedure.

27.2.2 Electrosurgery Techniques

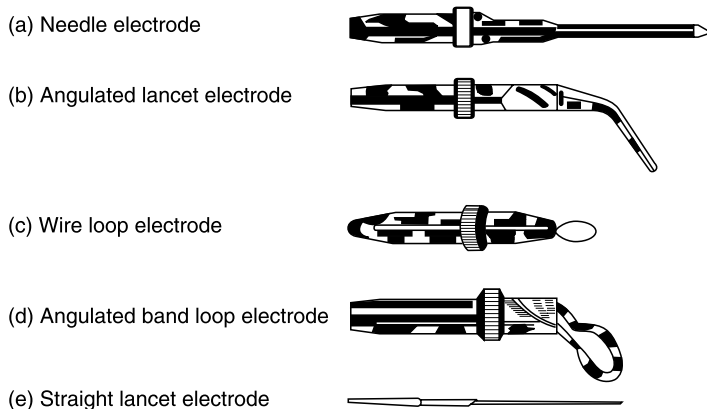
The electric current can flow only if the electric circuit is complete (closed). In terms of current flow, there are basically two types of electrosurgical techniques: the mono-polar and the bi-polar technique.

Mono-polar Technique: In the monopolar technique the current flows from the active electrode through the patient to the neutral electrode (patient plate) from which it returns to the generator. The cutting or coagulating effect depends on the contact area between the mono-polar active electrode and the tissue, which is very small compared with the contact area between the patient plate and patient's skin.

Bi-polar Technique: In this technique, two electrodes are used. The current in this case flows through the tissue between the tips of the two electrodes and returns to the generator without passage through the patient. The bipolar surgery is not only safer than monopolar but is also more precise since the current only flows locally at the specific site where it is actually required for heat generation. In addition, the risk of inadvertent burning of the patient at the patient plate is very low. Therefore, the bi-polar technique is becoming a method of choice wherever possible.

27.2.3 Electrodes Used with Surgical Diathermy

The bi-polar technique is used in most of the applications involving the use of surgical diathermy. The high potential terminal of the diathermy is connected to the cutting electrode which is mounted in an insulated handle. The cutting electrodes are available in a variety of shapes (Fig. 27.8), the choice depending upon the nature of application. Lancet electrodes are normally used for cutting applications whereas specific types of needle electrodes are preferred for epilation and desiccation. Loop electrodes are employed for exsecting (or opening up) channels and extirpating growths, etc. The active electrodes for coagulation purposes are of ball type (Fig. 27.9) or plate type. In electrosurgery, the surgeon must be able to switch the high frequency current on and off himself. This can be done with a finger-tip switch in the electrode handle or a foot switch.



► **Fig. 27.8** Cutting electrodes used with diathermy machine (a) needle electrode (b) angulated lancet electrode (c) wire loop electrode (d) angulated band loop electrode (e) straight lancet electrode

Consequently, the electrodes are used either with electrode handles with finger-tip switch, or with handles without finger-tip switch with the latter being used in conjunction with foot switches.

The low potential terminal of the radio frequency output leads is connected to the indifferent or dispersive electrode. This electrode consists of a lead plate (15 × 20 cm) wrapped in a cloth

bag, soaked in saline solution and strapped onto the patient's thigh. An alternative arrangement is to use a flexible non-crumpling stainless steel sheet plate without any covering. Good contact is established with the film of perspiration rising between the plate and the patient's body. Quite often, a liberal amount of conductive paste like ECG paste is applied to the plate. This gives excellent electrical contact and removes the need to keep a wet gauze pad. However, problems may arise if the paste is not cleaned from the plate after use as it may form a hard insulating layer.

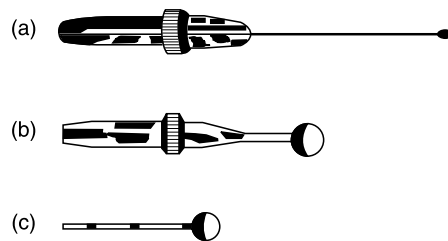
An alternative approach is to use capacitively coupled plates in which no direct contact is made between the metal of the indifferent electrode and the patient's skin. The electrode comprises a large sheet of thin metal sandwiched between two sheets of neoprene, which formed a capacitor with the patient's body. This capacitor allows an easy path for the passage of the high frequency diathermy currents. It, however, adds to the problem of introducing burn hazard by providing alternative current paths when other equipment with grounded patient connection is used.

Many manufacturers offer disposable indifferent electrodes and many advantages are claimed for these compared to conventional metal plates. One type of disposable dispersive electrode is like a disposable ECG electrode with a gel-soaked sponge backed by metal foil and surrounded by foam and pressure-sensitive adhesive.

There is no universal agreement over the safe effective area of the indifferent electrode necessary for diathermy current to exit without causing a rise in skin temperature. The American National Fire Protection Association manual (NFPA 1975) carries the suggestion that the plate area should be 1 cm^2 per 1.5 W applied power, leading to an area of 267 cm^2 for a 400 W unit. In Britain, the recommendation calls for 180 cm^2 plate area. Indifferent electrodes currently available range from 50 to 200 cm^2 .

The most common reason for faulty performance of an electrosurgical unit is improper placement of the indifferent electrode. This electrode must be placed in firm contact with a fleshy portion of the patient and as near as possible to the operating site. Poor contact or excessive distance from the operating site causes a considerable loss of energy available for the actual surgical procedure.

Equipments having rated output powers exceeding 50 W are generally equipped with a circuit that will interrupt the supply voltage to the equipment's output and sound an audible alarm whenever an interruption in the indifferent electrode circuit occurs. This is achieved by measuring the electrical impedance between different electrodes and the patient. If the high frequency rises above the limit permitted for the electrical impedance, an alarm will be generated and the output disconnected.



► **Fig. 27.9** Coagulation electrodes of different shapes and sizes

27.3 SAFETY ASPECTS IN ELECTROSURGICAL UNITS

The risks associated with electrosurgery fall into four main categories viz. burns, electrical interference with the heart muscles (ventricular fibrillation), the danger of explosions caused by sparks and electrical interference with pacemakers and other medical electronic equipment.

Burns: The predominant hazard associated with electrosurgical units is burns caused by excess current density at a rate other than that at which it is meant to be present. The burn usually

occurs at the dispersive electrode because of failure to achieve adequate contact. The injury can also occur because an unintended current pathway may be created. In the latter case, the lesion usually occurs at a point where the patient is inadvertently touching a grounded object and contact is made over a small area of skin.

The risk of burns also exists in the presence of moisture, i.e., the accumulation of prepping agents, blood or other fluids around the indifferent electrode can give rise to small, highly conductive areas. Burns resulting from small conductive areas between the limbs can be prevented by means of dry cloth placed between them.

During surgery, the output power of the electrosurgical unit should not be increased if the desired surgical effect is not obtained. Abnormal power settings indicate that something is wrong and that the fault must be identified. In particular, the indifferent electrode and all cables and connectors should be thoroughly checked. It is advisable to carry out surgical work with the power setting as low as possible, to reduce the risk of burns. Besides this, the active electrode, when not in use, should be placed well clear of the patient. This is to avoid its activation in case the foot switch is inadvertently pressed.

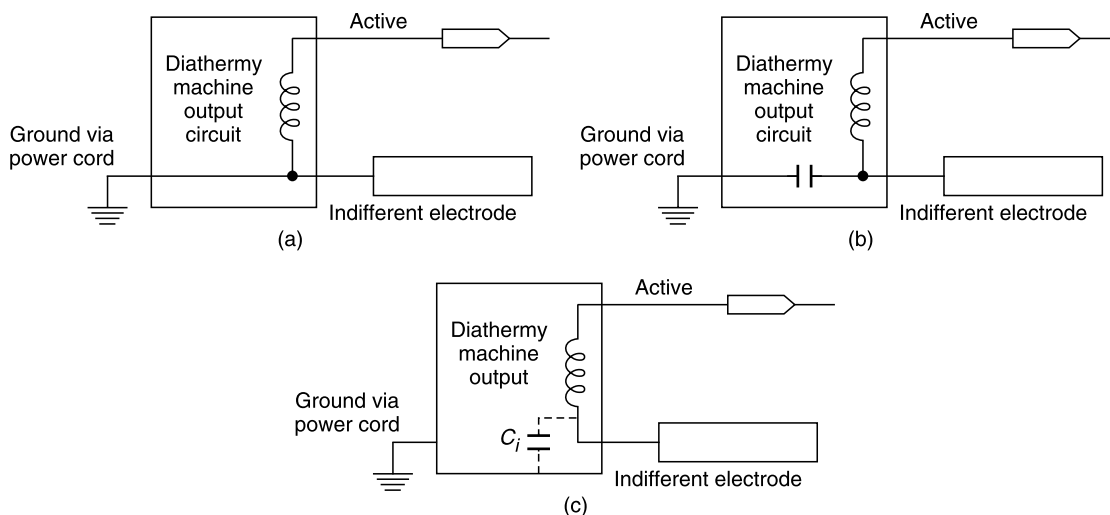
High Frequency Current Hazards: Another serious hazard associated with the use of surgical diathermy machines is the possible electrocution of the patient from faulty mains operated equipment, when one side of an electrical circuit is connected to earth. In order to provide protection against mains current electrocution, a capacitor (RF earthed) is generally included between the indifferent lead and earth.

The output configuration plays an important role in the RF current circuit. There are three technical approaches. In the earthed output system, the indifferent electrode is connected conductively to protective earth [Fig. 27.10(a)]. The earth referenced system uses a capacitor to connect the indifferent electrode to earth [Fig. 27.10(b)]. In the isolated system, the return electrode is floating, i.e., there is no intentional connection to earth [Fig. 27.10(c)]. The value of the capacitor is such that while providing a very low impedance to the high frequency diathermy current, it offers a higher impedance to the mains frequency. This approach also offers only a partial solution to a complex problem.

Modern solid-state machines usually have RF isolated patient circuits. This implies that ideally RF current may take only one path, i.e. from active electrode through the patient to the indifferent electrode. Since there is no earth connection, there is no propensity for the RF current to take any earth pathways which may unintentionally develop. However, due to RF leakage pathways inherent in the equipment and leads, no machine can be considered as completely isolated. The degree of RF leakage current is thus a measure of the degree of isolation of a particular machine. The lower the leakage current, the better the isolation. With the current technology, RF leakage figures of around 100 mA are generally achieved.

Of the three types of electrosurgical output systems—earthed, earth-referenced and isolated—only the last two are recommended by IEC (1978). For surgical applications in which the danger of ventricular fibrillation cannot be excluded, electrosurgical units of the isolated output type (type CF) should be used as they offer the best protection against fibrillation. Earth-referenced systems, type BF are recommended for most general applications.

The voltages of the power transformer in a surgical diathermy machine are high enough to cause serious injury. Therefore, when checking voltages, it is advisable to take adequate care. Also, caution should be taken to avoid damage to the test equipment due to high voltages and high currents.



► **Fig. 27.10** (a) Output circuit configuration in which the indifferent circuit is directly connected to the mains earth. (b) use of capacitor in the output circuit. This permits RF currents to flow to earth through the diathermy machine. However, it effectively blocks the passage of low frequency currents (50 Hz). (c) RF isolated output circuit configuration. Obviously, there is no direct connection of the indifferent circuit to ground. The RF leakage current is due to stray capacitance within the machine

Explosion Hazards: In operating theatres, danger zones can develop through the use of cleansing agents such as ether and alcohol, and by using explosive anaesthetic gas or mixtures with oxygen. The sparks associated with the use of surgical diathermy can cause a dangerous explosion if proper precautions are not taken. The use of non-explosive anaesthetics such as nitrous oxide, fluothane or halothane is recommended to prevent sparks generated during electrosurgery. If flammable gases are used, it is important that the electrosurgical unit be located outside the zone in which the anaesthetic is used. In addition, the foot-switches of the electrosurgical unit should always be explosion-proof.

Some diathermy machines are fitted with automatic anti-explosion devices which make the sparks occurring at the active electrodes innocuous. When the foot-switch is actuated or the finger-tip switch in the electrode handle is operated, this device causes a stream of nitrogen to emanate from the electrode handle to form a protective cloud around the cutting and coagulating electrode before the high frequency generator is switched on. Hence the explosive gas mixtures in the immediate vicinity of the electrode cannot ignite. An automatic control is incorporated in the unit which ensures that the high frequency current is not switched on until the active electrode is surrounded by protective gas. This is achieved by using an electrically heated thermistor in the handle which gets sufficiently cooled by the flow of protective gas. This ensures that an adequate stream of gas is emanating from the handle.

► 27.4 SURGICAL DIATHERMY ANALYSERS

The verification of output power of an electrosurgery unit is important to assure the continuing efficacy of the equipment and to promote consistency in the settings of output level controls.

This can be done with the help of a load consisting of non-inductive resistors and current metre suitable for the frequencies and waveform measured.

Electrosurgical devices are currently capable of delivering up to 500 W of power at fundamental output frequencies ranging from 0.3 to 3 MHz. In some units, significant harmonic components exist beyond 5 MHz (Wagner and Phillips, 1980). Even at these relatively low frequencies, precise measurements of output power require that adequate consideration be given to the radio-frequency nature of the output waveforms.

Fig. 27.11 shows the principle of operation of the electrosurgical analyser. It is known that the root-mean-square value of time varying repetitive current is given by:

$$I = \left[\frac{1}{T} \int_0^T [i(t)]^2 dt \right]^{1/2}$$

where I = RMS value

T = period of the current

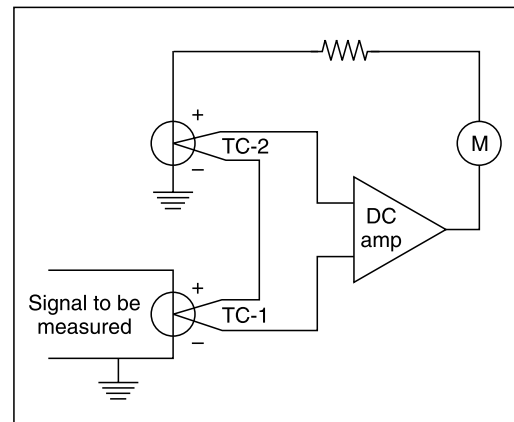
$i(t)$ = current as a function of time

If the repetitive time-varying current is a sinusoidal waveform, the relationship between the half-cycle average and the RMS value can be shown to be 1.1, and the RMS value can be determined by measuring the half-cycle average current and multiplying by 1.1.

A thermal converter consisting of a heater element, a thermocouple, and an evacuated glass envelope is used to measure the output power. The heater element varies in temperature as a function of the RMS current through it. The thermocouple is electrically insulated from, but thermally coupled to the middle of the heater element. It produces a voltage that is a function of the heater temperature. The evacuated glass envelope eliminates the loss of heat by convection and increases the sensitivity of the thermal converter.

The output of the thermal converter is proportional to the square of the applied voltage. This output when applied directly to a meter would give rise to a non-linear scale. Also, the readings are likely to vary with ambient temperature. By using two matched thermal converters, one heated by the signal to be measured and the other heated by dc signal, these problems can be conveniently overcome. The two thermocouple outputs are connected in series opposed. This combined signal is fed to the input of a high-gain DC amplifier. The output of the amplifier is fed back and drives the heater of the second thermal converter.

The operation of the system consists in applying the signal to be measured to the heater of the first thermal converter. This raises the temperature of the heater, and a DC voltage is generated by the thermocouple. This voltage is amplified by the amplifier and causes a current to flow in the heater of the second thermal converter. This, in turn, raises the temperature of the heater and



► Fig. 27.11 Principle of operation of surgical diathermy analyser

thermocouple. The voltage generated by the second thermocouple subtracts from the voltage of the first thermocouple and reduces the voltage applied to the amplifier. This process continues until the two heaters are at approximately the same temperature and the thermocouples are generating approximately the same voltage, thus balancing the system. Since the two thermal converters are a matched pair, and are at the same temperature, both heaters must have equal RMS currents flowing through them. The DC current flowing through the second thermal converter can be measured easily. Hence, the RMS value of the signal applied to the first thermal converter has also been measured.

■■■■► **MODEL QUESTIONS** ◀■■■■

1. Explain the principle of surgical diathermy. Illustrate the various types of waveforms used in surgical diathermy.
2. Draw the block diagram of a surgical diathermy machine. Why do we use isolated circuit in the output circuit?
3. What are the various types of electrodes used with surgical diathermy? Explain the difference between cutting and coagulating waveforms used in electrosurgery techniques.
4. What are the hazards associated with the use of electrosurgery units? How can high frequency circuit hazards be minimized? Explain with the help of a diagram.

Physiotherapy and Electrotherapy Equipment

29.1 HIGH FREQUENCY HEAT THERAPY

Physical stimulus commonly employed in the practice of physiotherapy is in the form of heat, either by simple heat radiation or by the application of high frequency energy obtained from special generators. The use of high frequency energy in thermotherapy has the advantage of considerable penetration as compared with 'simple' heat application. Thus, with high frequency energy, deeper lying tissues, e.g. muscles, bones, internal organs, etc. can be provided heat.

High frequency energy for heating is obtained by various ways. It may be from the short-wave therapy unit making use of either the condenser field or the inductor field method. Microwaves and ultrasonic waves are also used for heating purposes in special cases.

29.2 SHORT-WAVE DIATHERMY MACHINE

The term 'diathermy' means 'through heating' or producing deep heating directly in the tissues of the body. Externally applied sources of heat like hot towels, infrared lamps and electric heating pads often produce discomfort and skin burns long before adequate heat has penetrated to the deeper tissues. But with the diathermy technique, the subject's body becomes a part of the electrical circuit and the heat is produced within the body and not transferred through the skin (Yang and Wang, 1979).

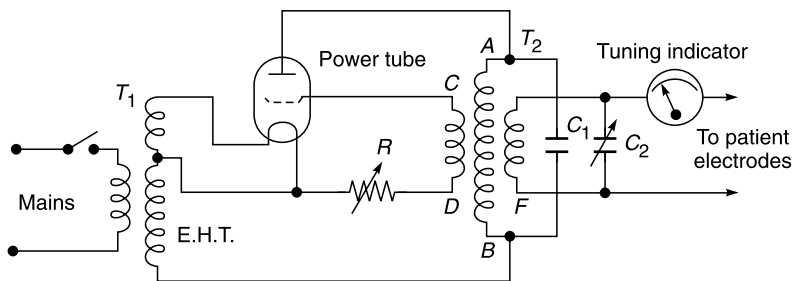
Another advantage of diathermy is that the treatment can be controlled precisely. Careful placement of the electrodes permits localization of the heat to the region that has to be treated. The amount of heat can be closely adjusted by means of circuit parameters. The heating of the tissues is carried out by high frequency alternating current which generally has a frequency of 27.12 MHz and a wavelength of 11 m. Currents of such high frequencies do not stimulate motor or sensory nerves, nor do they produce any muscle contraction. Thus, when such a current is passed through the body, no discomfort is caused to the subject. The current being alternating, it is possible to pass through the tissues currents of a much greater intensity to produce direct heating in the tissues similar to any other electrical conductor.

The method consists in applying the output of a radio frequency (RF) oscillator to a pair of electrodes which are positioned on the body over the region to be treated. The RF energy heats the tissues and promotes healing of injured tissues and inflammations.

Circuit Description: The short wave diathermy machine consists of two main circuits: an oscillating circuit, which produces a high frequency current and a patient circuit, which is

connected to the oscillating circuit and through which the electrical energy is transferred to the patient.

Earlier models of diathermy machines employed single-ended or push-pull power oscillators operating from unfiltered or partially filtered power supplies. They usually made use of a valve circuit, a typical example of which is shown in Fig. 29.1. Transformer T_1 , the primary of which can be energized from the mains supply, is a step-up transformer for providing EHT for the anode of the triode valve. A second winding can provide heating current for the cathode of the triode valve. The tank (resonance) circuit is formed by the coil AB in parallel with the condenser C_1 . The positive feedback is generated by coil CD. There is another coil EF and a variable condenser C_2 which form the patient's resonator circuit due to its coupling with the oscillator coil AB.



► Fig. 29.1 Simplified circuit diagram of a short wave diathermy unit

The anode supply of such a circuit is around 4000 V. The conduction in the triode takes place during the positive half-cycle and the high frequency is generated only during this period. More usually, the supply voltage is rectified before supplying to the anode of the oscillator valve. In such a case, the oscillations produced are continuous and more power thus becomes available. In order to ensure that the oscillator circuit and the patient's resonator circuit are tuned with each other, an ammeter is placed in series with the circuit. The variable condenser C_2 is adjusted to achieve a maximum reading on the meter, the needle swinging back on either side of the tuned position. The maximum power delivered by these machines is 500 W.

A thermal delay is normally incorporated in the anode supply which prevents the passage of current through this circuit until the filament of the valve attains adequate temperature. The patient circuit is then switched on followed by a steady increase of current through the patient. A mains filter is incorporated in the primary circuit to suppress interference produced by the diathermy unit itself.

There are several ways of regulating the intensity of current supplied to the patient from a short-wave diathermy machine. This can be done by either (i) controlling the anode voltage, or (ii) controlling the filament heating current, or (iii) adjusting the grid bias by change of grid leak resistance R_1 , or (iv) adjusting the position of the resonator coil with respect to the oscillator coil. However, the best way of finely regulating the current is by adjusting the grid bias, by putting a variable resistance as the grid leak resistance.

Automatic Tuning in Short-wave Diathermy Machines: Any short-wave therapy unit would give out the desired energy to the patient only if and as long as, the unit is correctly tuned to the electrical values of the object (part of the body). Therefore, tuning must be carefully carried out at the beginning of the treatment and continuously monitored during the treatment. There is

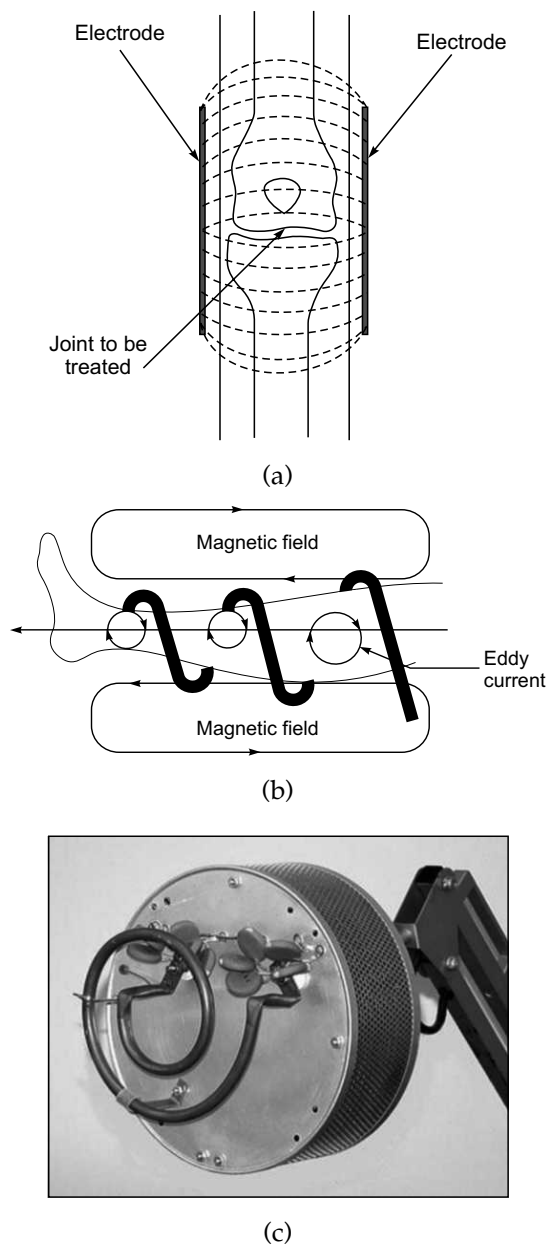
a possibility of the tuning getting affected due to unavoidable but involuntary movements of the patients and the resultant fall of dosage.

In order to overcome the problem of making tuning adjustments during the course of treatment, an additional circuit is fitted in the machine. The RF current in the patient circuit changes a capacitor to a voltage, whose polarity and magnitude is a measure of the detuning of the patient circuit. This voltage accordingly moves a servo-motor, adjusting the tuning capacitor so that resonance is restored.

Application Techniques of Short-wave Therapy: The pattern of tissue heating is greatly affected by the method of short-wave diathermy delivery. The two most common forms of application include the capacitor plate method and the inductive method (Fig. 29.2).

In the capacitor plate method, the output of the short-wave diathermy machine is connected to metal electrodes which are positioned on the body over the region to be treated (Fig. 29.3). These electrodes are called 'PADS' in the terminology of the diathermy. These pads or electrodes do not directly come into contact with the skin. Usually layers of towels are interposed between the metal and the surface of the body. The pads are placed so that the portion of the body to be treated is sandwiched between them. This arrangement is called the 'Condenser Method' [Fig. 29.2(a)] wherein the metal pads act as two plates while the body tissues between the pads as 'dielectric' of the capacitor. When the radio frequency output is applied to the pads, the dielectric losses of the capacitor manifest themselves as heat in the intervening tissues. The dielectric losses may be due to vibration of ions and rotation of dipoles in the tissue fluids (electrolytes) and molecular distortion in tissues which are virtually insulators like fats.

There are two types of electrode arrangements for condenser method. In the *Contraplanar technique*, the electrodes are placed over the opposite aspects of the trunk or limb, so that the electric field is directed through the deep tissues. On the other hand, in the *coplanar technique*, the electrodes are placed



► **Fig. 29.2** Methods of applying electrodes in shortwave diathermy treatment (a) condenser method (b) inductive method (c) Inductive heating by a coil housed in a drum.

side by side on the same aspect of the part, provided there is an adequate distance between them. Fig. 29.3 shows a shortwave diathermy in use with coplanar electrodes.

Alternatively, the output of the diathermy machine may be connected to a flexible cable instead of pads. This cable is coiled around the arm [Fig. 29.2(b)] or knee or any other portion of the patient's body where plate electrodes are inconvenient to use. When RF current is passed through such a cable, an electrostatic field is set up between its ends and a magnetic field around its centre. Deep heating in the tissue results from electrostatic action whereas the heating of the superficial tissues is obtained by eddy currents set up by a magnetic effect. This technique is known as 'inductothermy'.

Another form of inductive heating is by a coil which is housed within a drum. The current flowing within the coil produces a rotating magnetic field, which in turn produces eddy currents in the tissues. Due to the friction caused by the Eddy currents, heat is produced in the tissues. The configuration of the coil is shown in Fig. 29.2 (c).

Although most short-wave diathermy machines have an output power control, yet there is no indication of the amount of converted and absorbed heat within the body tissues. Therefore, the intensity of treatment is dependent on the subjective sensation of warmth felt by the patient.



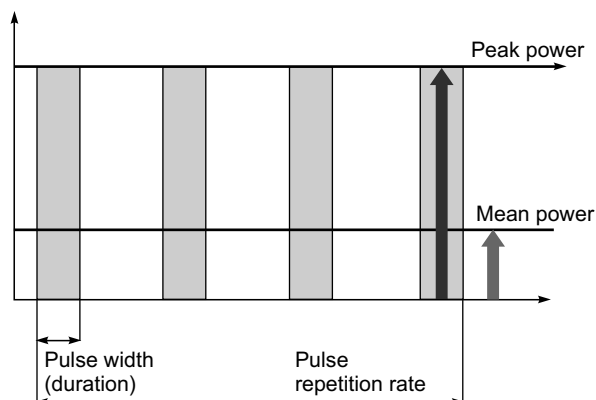
► Fig. 29.3 Shortwave diathermy (with capacitive electrodes) in use (<http://www.holistichealthpc.com/services/procedures/>)

29.2.1 Pulsed Shortwave Therapy

A severe limitation of diathermic machines is that they direct continuous high frequency radio waves, and if a high enough output of energy is sustained for even a brief time, they can cause burns. As a consequence, the wattage has to be lowered to tolerable limits. Also, the heat resulting from diathermy has many contra-indications and limitations because the heat limits the amount of energy that can be used. The increase in energy output, while avoiding the dangers of heat, has been achieved in a machine called 'Pulsed Shortwave Therapy' (PSWT).

Pulsed therapy apparatus also works at 27.12 MHz, the frequency of short-wave diathermy machine. However, the energy is delivered in the form of pulses of 65 ms with an interval between pulsations at a maximum setting of 1600 ms. The rate of pulsations is adjustable in steps from 80 to 600 pulses per second. Even at this setting, power is provided no more than 4% of the total time that the machine is in operation. The peak instantaneous wattage can be varied from 290 to 975 W. The effect of the rest periods is to reduce the output to a maximum average of only 40 W. The result is an intermittent, relatively athermic, electrotherapy (Fig. 29.4). Therefore, despite the high energy pulses, all the heat is dissipated during the rest period, and as a consequence, there is no danger of burns or hyperthermic complications. The depth of penetration depends upon the peak energy delivered, which is adjustable on the machine. Tuning adjustment is provided to obtain maximum efficiency at each wattage setting. The control offered by the machine will enable the user to vary (a) the mean power delivered to the patient and (b) the pulsing parameters governing the mode of delivery of the energy. Studies show that *mean power* is probably the most important parameter for treatment.

There are two basic types of output from these machines, the *electric* field, comparable to the condenser (capacitor) field in traditional shortwave diathermy and secondly, the *magnetic* field,



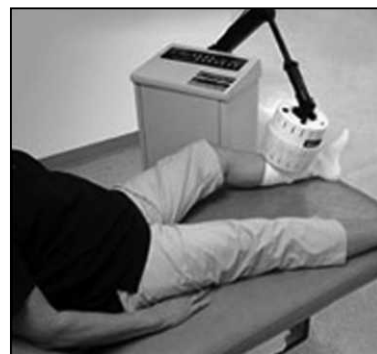
► Fig. 29.4 Peak power vs mean power in a pulsed shortwave therapy machine

comparable to inductothermy. Some machines offer the facility to pulse either output, with the magnetic field being delivered via a drum containing a coiled conductor housed in some form of 'monode' or 'drum' applicator (Fig. 29.5). Manufacturers include a special screen in the face of the drum to eliminate most, if not all of the electric field.

The output of the monode applicator can be thought of as a form of pulsed inductothermy. The pulsed electromagnetic field which is emitted from the applicator will be transmitted through the tissues, and will be absorbed in those of low impedance i.e. the conductors which are tissues like muscle, nerve, those which are highly vascular, tissues in which there is oedema, effusion or recent haematoma.

The mean power table that is provided with each machine is the critical piece of information when it comes to clinical decision making and patient doses. The table identifies all the potential combinations of pulse repetition rate and pulse durations, and therefore how the machine can be 'set' to deliver a specific mean power. The tables are not interchangeable between models and it is important that the correct table is used for the machine available or else an incorrect clinical dose could be applied.

The physiological effects of pulsed therapy are not fully understood. Cells *in vitro* become aligned within a diapulse field as iron filings follow the lines of force around a magnet, and the orientating effects of the electromagnetic waves seem to be important. Perhaps the most exciting possibility with pulsed-electromagnetic waves is that they may enhance the rate at which peripheral nerves, particularly the smaller diameter fibres, re-generate.



► Fig. 29.5 Pulsed shortwave diathermy machine with monode applicator

29.3 MICROWAVE DIATHERMY

Microwave diathermy consists in irradiating the tissues of the patient's body with very short wireless waves having frequency in the microwave region. Microwaves are a form of electromagnetic radiation with a frequency range of 300-30,000 MHz and wavelengths varying

from 10 mm to 1 m. In the electromagnetic spectrum, microwaves lie between short waves and infrared waves. The most commonly used microwave frequency for therapeutic heating is 2450 MHz corresponding to a wavelength of 12.25 cm. The heating effect is produced by the absorption of the microwaves in the region of the body under treatment.

Microwave diathermy provides one of the most valuable sources of therapeutic heat available to the physician. However, in many conditions, though the therapeutic effects of microwave diathermy are similar to short-wave diathermy, yet in others, better results are obtained by using microwave.

The technique of application of microwave diathermy is very simple. Unlike the short-wave diathermy where pads are used to bring in the patient as a part of the circuit, the microwaves are transmitted from an emitter, and are directed towards the portion of the body to be treated. Thus, no tuning is necessary for individual treatments. These waves pass through the intervening air space and are absorbed by the surface of the body producing the heating effect. The special design of the treatment heads and shapes focuses the field directly at the target area. The whole device is used to direct the waves onto the tissues. This device is sometimes termed the emitter, the director or the applicator. In this kind of treatment, the patient does not form a part of the circuit, so no tuning is necessary as in short wave application. Generally, two designs of applicators are common:

- (i) **Circular:** In which heat is denser at the periphery more than at the center. Different sized applicators are used for small or large areas.
- (ii) **Rectangular:** for elongated areas.

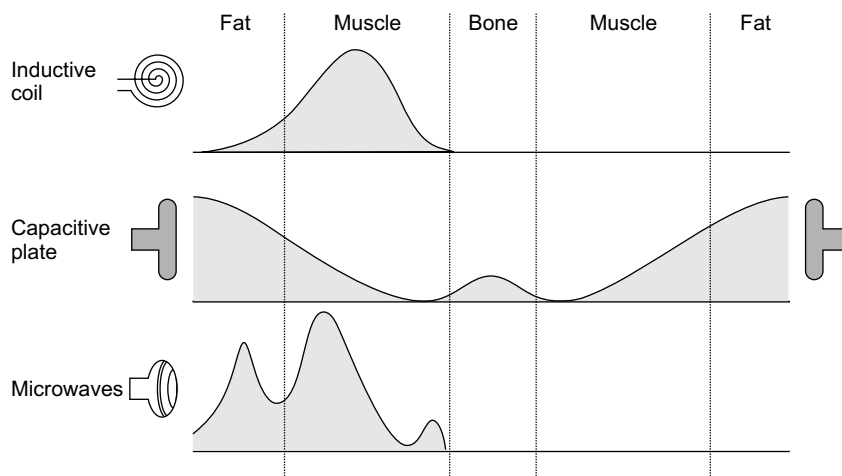
So, the applicator selection depends upon the shape and the anatomical configuration of the treated area.

Microwave penetrates more deeply than infrared rays but not deeply as shortwave diathermy, so microwave is not suitable for deeply placed structures (Cameron, 2009). The effective depth of microwave penetration is about 3 cm so the depth of heating is intermediate between that of infrared radiation and short wave diathermy. As the microwaves are strongly absorbed by water, tissues with high fluid content are heated most, while less heat is produced for tissues with low fluid content as fat. Moreover, there is an appreciable heating of tissues having good blood supply such as muscle. Fig. 29.6 shows comparison of heat distribution in the body-tissues with shortwave and microwave diathermy.

Production of Microwaves: Microwaves are produced by high-frequency currents and have the same frequency as the currents which produce them. Such currents cannot be produced with oscillators using ordinary vacuum tube valves or solid-state devices. A special type of device called 'magnetron' is used for the production of high frequency currents of high power.

The magnetron consists of a cylindrical cathode surrounded by an anode structure that contains cavities opening into the cathode-anode space by means of slots. The output energy is derived from the resonator system by means of a coupling loop which is forced into one of the cavities. The energy picked up on the coupling loop is carried out of the magnetron on the central conductor of a co-axial output tube through a glass seal to a director. The director consists of a radiating element of antenna and a reflector which directs the energy for application to the patient. The electrical current is transformed into electromagnetic radiation on passing through the antenna.

The reflector then focuses this electromagnetic energy and beams it to the tissues where it is subsequently absorbed, reflected or refracted, according to the electrical properties of the tissue. Tissues of lower water content (i.e., subcutaneous) are penetrated to a greater depth but little is



► **Fig. 29.6** Comparison of the heat distribution in the body tissues with the shortwave inductive diathermy, capacitive plate of diathermy applicator and microwave diathermy. (Adapted from Cameron, 2009)

absorbed, whereas tissues of high water content (i.e., muscle) absorb more of the electromagnetic energy but allow little penetration.

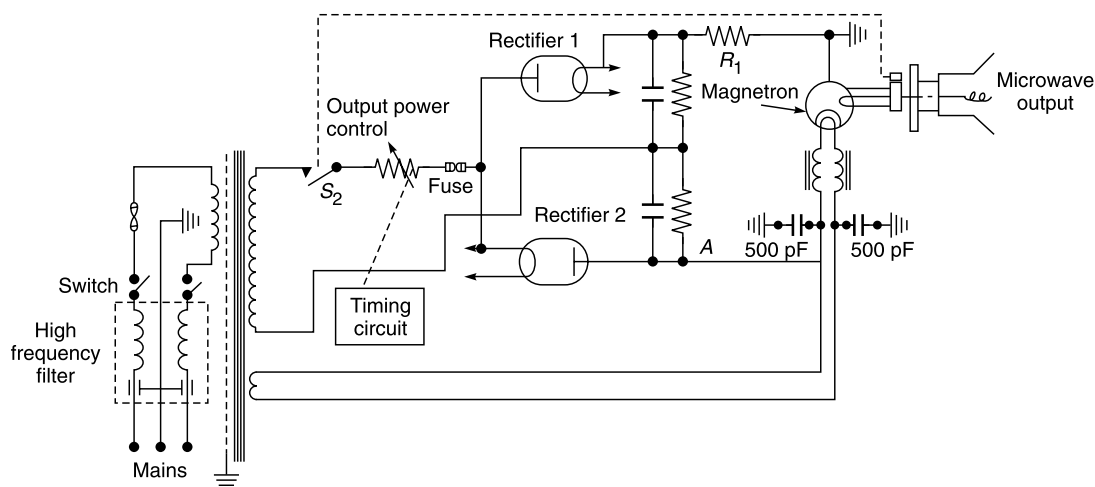
The output power of a magnetron depends upon anode voltage, magnetic field and the magnitude and phase of the load impedance to which the magnetron output power is delivered. Therefore, the cable used to carry the energy from the magnetron to the director is always of a definite length for a particular frequency. A part of the energy fed to the magnetron is also converted into heat in the anode on account of the collision of the electrons with the anode so that the output energy is considerably less than the input energy. The efficiency of a magnetron is usually 40 to 60%. The heat produced at the anode must be removed which is usually done by using water or air as a means of cooling.

Schematic Diagram of a Microwave Diathermy Unit: The essential parts of a microwave diathermy unit are shown in Fig. 29.7. The mains supply voltage is applied to an interference suppression filter. This filter helps to bypass the high frequency pick-up generated by the magnetron. A fan motor is directly connected to the mains supply. The fan is used to cool the magnetron.

The Delay Circuit: It is necessary for the magnetron to warm up for 3 to 4 minutes before power may be derived from it. A delay circuit is incorporated in the apparatus which connects the anode supply to the magnetron only after this time elapses. The arrangement is such that a lamp lights up after 4 minutes indicating that the apparatus is ready for use.

The Magnetron Circuit: The magnetron filament heating voltage is obtained directly from a separate secondary winding of the transformer. The filament cathode circuit contains interference-suppression filters. The anode supply to the magnetron can be either DC or AC. A DC voltage is obtained by a full wave rectifier followed by a voltage doubler circuit. A high wattage variable resistance is connected in series which controls the current applied to the anode of the magnetron.

When using AC, the voltage is applied to the anode of the magnetron through a series connected thyatron so that the AC voltages of both tubes are equal in phase. By shifting the



► Fig. 29.7 Simplified circuit diagram of a microwave diathermy machine

phase of the control grid voltage with respect to the phase of the anode voltage, the amount of current through the magnetron can be determined and thus the output power can be varied. The phase shift can be achieved by using a capacitor resistor network.

Safety Circuits: There are chances of the magnetron being damaged due to an excessive flow of current. It is thus protected by inserting a fuse (500 mA) in the anode supply circuit of the magnetron. The protection of both the patient and the radiator is ensured by the automatic selection of the control range depending on the type of the radiator used.

The considerable interference produced by the apparatus necessitates the use of large self-inductance coils in the primary supply. Since the cores are likely to become saturated in view of the small dimensions, the coils are split up and fitted in such a way that no magnetization occurs.

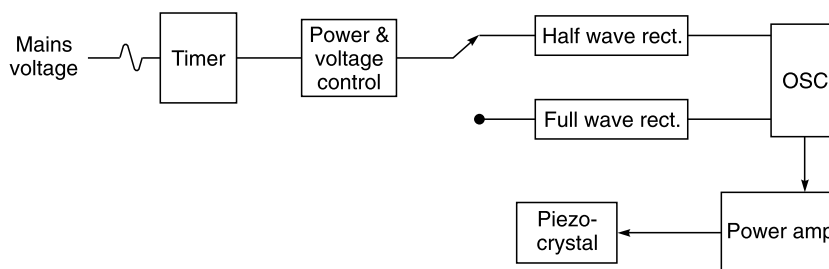
Excessive dosage can cause skin burns and in all cases, the sensation experienced by the patient is the primary guide for application. The skin should be dry as these waves are rapidly absorbed by water. The duration of irradiation generally ranges from 10 to 25 minutes.

■ 29.4 ULTRASONIC THERAPY UNIT

Ultrasonics are used for therapeutic purposes in the same manner as a short-wave diathermy machine is used. The heating effect in this case is produced because of the ultrasonic energy absorption property of the tissues. The property of specific heat distribution in tissue and the additional effect of a mechanical component have given rise to a number of special therapeutic applications of ultrasonics. The effect of ultrasonics on the tissues is thus a high speed vibration of micro-massage. Massage as a modality in physical medicine has been used in the treatment of soft tissue lesions for centuries. Ultrasonic energy enables this massage to be carried out, firstly to a greater depth than is possible manually, and secondly at times (in acute injuries) when pressure cannot be exerted by hand because of intolerable pain caused to the patient. The thermal effects of ultrasound are dependent on the amount of energy absorbed, the length of time of the ultrasound application and the frequency of the ultrasound generator. The electrical power required in most of the applications is usually less than 3 W/cm^2 of the transducer area that is in contact with the part of the body to be treated.

Ultrasonic generators are constructed on the piezo-electric effect. A high-frequency alternating current (e.g., 0.75-3.0 MHz) is applied to a crystal whose acoustic vibration causes the mechanical vibration of a transducer head, which itself is located directly in front of the crystal. These mechanical vibrations then pass through a metal cap and into the body tissue through a coupling medium. The therapeutic ultrasonic intensity varies from 0.5 to 3.0 W/cm². Applicators range from 70 to 130 mm in diameter. The larger the diameter of the applicator, the smaller would be the angle of divergence of the beam and the less the degree of penetration.

Circuit Description: The equipment required for ultrasonic therapy is electronically very simple. Fig. 29.8 shows the block diagram. The heart of the system is a timed oscillator which produces the electrical oscillations of the required frequency. The oscillator output is given to a power amplifier which drives the piezo-electric crystal to generate ultrasound waves. Power amplification is achieved by replacing the transistor in typical LC tuned Colpitt oscillator by four power transistors placed in a bridge configuration.



► Fig. 29.8 Block diagram of an ultrasonic therapy unit

The delivery of ultrasound power to the patient is to be done for a given time. This is controlled by incorporating a timer to switch on the circuit. The timer can be a mechanical spring-loaded type or an electronic one, allowing time settings from 0 to 30 minutes.

The output of the oscillator can be controlled by either of the following two methods:

- Using a transformer with a primary winding having multi-tapped windings and switching the same as per requirement;
- Controlling the firing angle of a triac placed in the primary circuit of the transformer, and thereby varying the output of the transformer.

The machine can be operated in either continuous or pulsed mode. A full-wave rectifier comes in the circuit for continuous operation. The mains supply is given to the oscillator without any filtering. The supply voltage is therefore at 100 Hz which causes the output 1 MHz to be amplitude modulated by this 100 Hz. In pulsed mode, the oscillator supply is provided by the half-wave rectifier and the oscillator gets the supply only for a half cycle. Thus the output 1 MHz is produced only for one half of the cycle and is pulsed.

The transducer is a lead zirconate titanate crystal, having 5-6 cm² effective radiating area. The length of the cable connecting the transducer with the oscillator is of critical dimension and should not be altered. In front of the crystal lies a metal face plate which is made to vibrate by the oscillations of the crystal. Ultrasonic waves are emitted from this plate. The crystal has a metal electrode pressed against its back surface by a coiled spring. Voltage is applied to the crystal via this electrode. The front diaphragm is grounded and provides a return path for the excitation voltage.

Dosage Control: The dosage can be controlled by varying any of the following variables.

- Frequency of ultrasound;
- Intensity of ultrasound; or
- Duration of the exposure.

The frequency is involved because the absorption of ultrasonics by the tissues is a frequency-dependent phenomenon. The question as to which ultrasonic frequency to use has been the subject of much investigation and thought but it has been established that a frequency of approximately 1 MHz is the most useful. The amount of energy absorption in the human tissue has been measured experimentally and in soft tissue, a reduction of 50% occurs with a 1 MHz ultrasonic transmission at a depth of 5 cm. The higher the frequency the quicker the energy loss and thus with a transmission of 3 MHz, this reduction of 50% occurs at a depth of only 1.5 cm. Below a frequency of 1 MHz, the beam of ultrasonic energy tends to diffuse and no efficient treatment can be expected. A frequency in the range of 800 kHz to 1 MHz is, therefore, most widely adopted.

Unlike the operation of a short-wave therapy unit, no tuning is necessary while the treatment is in progress. The operating frequency is also not very critical and may vary to the extent $\pm 10\%$.

The output power of an ultrasonic therapy unit can be varied continuously between 0 and 3 W/cm^2 . The calibrated positions are marked in steps. The steps indicate the average value of intensity monitored in terms of electric power converted into acoustic energy.

Standard values concerning dosing in ultrasonic therapy have been established on the basis of experimental studies. In order to achieve maximum therapeutic efficiency, it is necessary to ascertain the correct ultrasonic intensity and duration of application for a given indication. To make matters simpler, some instruments are equipped with a 'dose tabulator' from which the data concerning dosing can be taken at a glance. In this table, a dose mark is given for every indication (disease) and all that is required is to set a pointer appropriately to ensure that the apparatus is providing the correct output intensity.

Apart from continuous and mains frequency modulated modes, ultrasonic output can also be modulated by any other frequency. The idea behind a pulsed operation is that the predominant effect of ultrasound is not the heating effect but the direct mechanical effect (micro-massage). The thermal effect is reduced by repeatedly interrupting the supply of energy through brief pauses.

Application Technique: There are several ways for applying ultrasonics to the body. The probe can be put in direct contact with the body (Fig. 29.9) through a couplant provided the part to be treated is sufficiently smooth and uninjured. In case a long area is to be treated, the probe is moved up and down, and for small areas it is given a circular motion to obtain a uniform distribution of ultrasonic energy. If there is a wound or an uneven part (joints etc.), the treatment may be carried out in a water bath. This is to avoid mechanical contact with the tissues which may damage an already injured surface. It should be ensured that air bubbles are not present either on the probe or the skin.



► Fig. 29.9 Ultrasonic therapy in use
(Courtesy: M/s Chattanooga, U.S.A.)

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Cryotherapy

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Continuing Education Activity

Cryosurgery was first described in the 1800s and has since evolved into a mainstay of therapy within dermatology. Cryosurgery is an effective alternative to more invasive techniques and can be delivered quickly and cost-effectively in an outpatient setting. Because this is a non-invasive technique, cryosurgery can produce excellent cosmetic outcomes. This activity highlights the indications, contraindications and complications of cryotherapy.

Objectives:

- Describe the indications for cryotherapy.
- Summarize the complications of cryotherapy.
- Review the mode of action of cryotherapy.
- Summarize the importance of improving care coordination among interprofessional team members to improve outcomes for patients undergoing cryotherapy.

[Access free multiple choice questions on this topic.](#)

Introduction

Cryosurgery was first described in the 1800s and has since evolved into a mainstay of therapy within dermatology. Cryosurgery is an effective alternative to more invasive techniques and can be delivered quickly and cost-effectively in an outpatient setting. Because this is a non-invasive technique, cryosurgery can produce excellent cosmetic outcomes. Cryosurgery is performed using a cryogen, typically liquid nitrogen, to cool the targeted tissue to subzero temperatures. This effect induces tissue damage in 2 mechanisms. The first mechanism is the induction of tissue ischemia by damaging blood vessels and capillaries within the target area which leads to ischemic necrosis of the tissue. The second mechanism damages the cells more complexly by forming ice crystals and inducing osmotic cell injury and cellular membrane disruption. As the tissue is cooled, ice crystals form between cells which creates an osmotic gradient rapidly drawing water out of the cells. As cooling continues, crystals form within the cell which can lead to rupture of the cell. The thawing process also damages cells. As the tissue thaws, crystals outside the cells melt which creates a gradient that rapidly draws water back into the cell which can cause cells to swell and burst.[2][3][4][3][2]

There is an advantage to cryosurgery compared to excisional techniques for malignancies. When a malignant lesion is excised, the host is no longer exposed to the antigens present in the malignant cells. In the case of cryosurgery, however, the antigens present on the dead malignant cells are retained, allowing for a host immune response that may lead to a systemic response to the malignant cells. [5]

The amount of tissue damage increases with each freeze thaw cycle. The target temperature for the destruction of benign cells is -20 C. Cancerous cells can be more resistant to cell death and require -50 C to be destroyed. Unfortunately, melanocytes are very susceptible to thermal injury and may die at temperatures below -5 C. [4]

Zones of spherical freezing advance are known as isotherms. The radius for each isotherm is the same in all directions. For example, a -5 C degree isotherm may have a 10 mm distance away from the center of freezing in both depths and along the surface of the skin. A -10-C isotherm may be found at 5 mm away from the center and represent a temperature of -10 C at 5 mm in depth. Knowing the target temperature and understanding the concept of isotherms, a cyrosurgeon can achieve a targeted temperature for a certain depth by measuring the temperature at the periphery of his zone of freezing. [1]

It is also important to understand that conduction will play a role in the temperature of a lesion. Air is the least conductive material for heat, so spraying a lesion with liquid nitrogen from a distance will be less effective compared to holding the spray tip close to the lesion. Keratin is also a poor conductor so for hyperkeratotic lesion it may be prudent to debulk the lesion before performing cryosurgery.

Indications

Cryosurgery has a number of indications for both malignant and benign lesions.

Benign lesions that can be treated with cryosurgery include seborrheic keratosis, verruca, skin tags, molluscum contagiosum, solar lentigo and hypertrophic/keloid scars. Most of the entities can be treated with a single round of cryosurgery, but for larger or thicker lesions treatments can be repeated at 3 to 4-week intervals until the lesions have resolved. This is especially true in verruca which typically take anywhere from 2 to 6 treatments to resolve.

Pre-malignant and malignant lesions that can be treated with cryosurgery include actinic keratosis, basal cell carcinoma, and non invasive squamous cell carcinoma. Cryosurgery has been used to treat lentigo maligna melanoma with variable efficacy and recurrence rates. Actinic keratosis is commonly treated with cryosurgery. However, the treatment of malignant lesions with cryotherapy is not a first-line therapy and is typically reserved for patients who are not good candidates for excision. Large lesions that would be very disfiguring if excised or elderly patient who cannot tolerate excision are examples where cryosurgery may be indicated. [6][7][8]

Contraindications

Cryosurgery should not be performed on a neoplasm of uncertain behavior. Before cryosurgery is performed a diagnosis must be made either by histologic or clinical and dermatoscopic diagnosis.

Other contraindications for cryosurgery are conditions that can be exacerbated by cold exposure such as cryoglobulinemia, multiple myeloma, Raynaud disease, cold urticaria, previous history of cold induce injury at the site or limb, as well as poor circulation at the site or in that limb. Cryosurgery can induce vasoconstriction so cryosurgery to a vascularly impaired area may lead to undesired tissue necrosis.

Equipment

There several different cryogens including nitrogen gas, carbon dioxide gas, and other compressed gases, but the most common cryogen by far is liquid nitrogen. The boiling point of liquid nitrogen is -196 C. Liquid nitrogen is typically stored in heavily insulated long-term storage tanks called dewars. Dewars can range in size from 4 to 50 L and can be used to store liquid nitrogen for up to 2 months. Devices that generate liquid nitrogen are also available if needed.

Cryosurgical devices are 300 to 500 mL units can temporarily hold liquid nitrogen and typically come with a number of different attachments based on the desired treatment technique. Several different treatment approaches will be discussed below, but the equipment necessary for those treatments will be discussed here. For open techniques that require spraying liquid nitrogen, there are cryosurgical units that come with a spray gun-type design that comes with a variety of different sized tips that can be used based on the size of the treatment area. Cones that look like an otoscope tip and plates of various-sized cutouts can provide more targeted delivery of liquid nitrogen. There are also variously sized probes that can be attached to the cryosurgical units to when direct contact treatment is desired. In addition, forceps or needle drivers can be dipped into the cryosurgical units and frozen for approximately 20 to 30 seconds and then used to grasp pedunculated lesions.

If treatment of malignant lesions is performed, it is advisable to use a device to determine the temperature of the tissue. For probes, a thermometer may be built into the probe and can, therefore, give the physician an idea of the

temperature at the tip of the probe. There are also infrared thermometers that can be used in conjunction with spray cans to give the physician an idea of the temperature at different isotherms.

Preparation

Minimal skin preparation is needed for cryosurgery. Antiseptics are not typically indicated in a majority of cryosurgeries. However, for treatment of malignant lesions, especially with a contact probe, topical antiseptics should be applied since there is a risk of bleeding. For hyperkeratotic lesions, curettage should be performed to allow for better and more accurate administration of liquid nitrogen. For small lesions, injection of local anesthetic may be more painful than the cryosurgery itself and is therefore not indicated. In general, cryosurgery of small lesions is well tolerated. For larger areas, topical anesthetics can be applied several hours before the procedure to help reduce the pain associated with freezing. Interestingly, anesthetics may be less effective for pain experienced during the thawing period.

Technique

Cryosurgical techniques vary based on the lesion being treated. Benign lesions can typically be treated with a single freeze-thaw cycle with a target temperature of -25 C for keratinocytic tumors and -5 C for pigmented lesions. Whereas, malignant lesions are typically treated with 2 cycles and a target temperature of -50 C. An exception to benign lesions receiving one freeze-thaw cycle is verruca which due to there relative resistance to treatment can be treated with 2 freeze-thaw cycles at the discretion of the surgeon[9]. Thawing is not complete until the entire lesion has returned to its original color and frost is no longer observed. Allowing the lesion to thaw completely produces maximum tissue destruction since thawing also produces tissue damage. As a general rule for benign lesions, it is better to err on the side of caution and undertreat lesions since any persistence can be treated at a later visit. This approach can prevent unnecessary complications such as hypopigmentation. Benign lesions should be frozen until the white halo extends 1 to 2 mm beyond the border of the lesion to guarantee that the entire lesion is removed. The duration of freezing for solar lentigo are typically around 3 to 4 seconds since melanocytes are more sensitive to cold injury, whereas benign lesions are typically frozen for 5 to 10 seconds. The margins for a solar lentigo should be just beyond the border of the lesion to prevent persistent pigment at the margins of the lesion.

Although cryosurgery is the first-line treatment for actinic keratosis, cryosurgery is not a first-line therapy for malignant lesions. Treatment of actinic keratosis or malignant lesions is typically performed with 2 freeze-thaw cycles with the size of the margin dependent on the malignancy. Actinic keratosis is typically frozen for approximately 10 to 15 seconds with 2 mm margin. The target temperature for malignant lesions is -50 degrees for each of the 2 freeze-thaw cycles. It typically takes approximately 40 to 90 seconds to achieve these temperatures. In general, nonmelanoma skin cancers should have a margin of at least 5 mm. There have been reports of melanoma in situ being treated with cryosurgery with a margin of 10 mm[10]. Recurrences are difficult to identify and recurrence rates range from 6 % to 34%. In contrast to benign lesions, cryosurgery of malignant lesions should err on the side of overtreatment and be more aggressive because margin control is not possible with cryosurgery. Since there is no excised tissue to be examined following this procedure, there is no way definitive option to ensure that margins have been cleared, which may lead to the possibility of recurrence. For this reason, caution should be used with this approach, and regular follow-up is recommended to monitor for recurrence. [1][11]

There are several different techniques for the administration of liquid nitrogen. The most common cryosurgical technique is the open technique in which liquid nitrogen is sprayed onto the target lesion. Intermittent release of liquid nitrogen provides a more controlled and precise application compared with spraying in a continuous fashion. The semi-open technique is best for papular lesions. In this technique, a cone or plate covers or directs liquid nitrogen to the target area and allows for a more targeted treatment area. The closed/contact technique uses a probe cooled with liquid nitrogen applied directly to the skin. Probes may freeze to the skin during the procedure and forceful removal of the probe can rip tissue off of the treatment area and lead to bleeding. To prevent this, the probe should be frozen prior to touching the skin. In the event that the probe does become frozen to the skin, it should be gently warmed or warm water can be applied to release the probe. Another form of the close/contact technique is the use of liquid nitrogen to cool the tips of forceps or needle drivers, which are then used to grasp pedunculated lesions.

Complications

There are a number of expected outcomes following cryosurgery that patient should be advised about before the surgery. Cryosurgery produces tissue damage that heals by secondary intention, which can take longer to heal than an excision, especially if it is performed on the leg. Healing time directly correlates with the depth of freezing, so lesions that have had a deeper freeze will take longer to heal. Pain is an expected outcome of this procedure. Pain will last less than a minute. Treated sites will progress from erythema, edema, and vesiculation over the course of several days. Depending on the depth of the treatment an exudate may be noted up to 2 weeks after treatment. Mummification or eschar formation may follow the exudative phase and can be gently debrided if necessary.

Complications of cryosurgery include dyspigmentation, alopecia, pseudoepitheliomatous hyperplasia, depressed scars, and tissue distortion (nail dystrophy or notching of cartilage). Dyspigmentation is the most common complication of cryosurgery. Hypopigmentation is the most likely type of dyspigmentation seen because cryosurgery can destroy melanocytes. However, darker skinned individuals can develop hyperpigmentation. In the scalp or other hair-bearing areas, alopecia can result from cold-induced destruction of hair bulge cells and lead to permanent alopecia. Pseudoepitheliomatous hyperplasia can result from cryotherapy. No treatment is needed as this will spontaneously resolve. Depressed scars can form after deep cryosurgery. The depressions may resolve over time. Cryosurgery can also damage underlying structures such as the nail matrix of the digits or cartilage on the ears and nose, which can produce notching or retraction.

Clinical Significance

Cryosurgery is a common dermatologic procedure that can be used for a number of benign and malignant conditions. It is important for any physician performing cryosurgery to understand the mechanism of action and application of cryosurgical techniques for treating these techniques. Physicians should use their knowledge of the uses and contraindications of cryosurgery to select appropriate candidates for this treatment. Patients should be counseled on the expected outcomes and possible complications that may arise due to cryosurgery.

Enhancing Healthcare Team Outcomes

Cryotherapy is performed by many healthcare professionals. However, all providers need to know the potential complications and warn the patient. There are a number of expected outcomes following cryosurgery that patient should be advised about before the surgery. Cryosurgery produces tissue damage that heals by secondary intention, which can take longer to heal than an excision, especially if it is performed on the leg. Healing time directly correlates with the depth of freezing, so lesions that have had a deeper freeze will take longer to heal. Pain is an expected outcome of this procedure. Pain will last less than a minute. Treated sites will progress from erythema, edema, and vesiculation over the course of several days. Depending on the depth of the treatment an exudate may be noted up to 2 weeks after treatment. Mummification or eschar formation may follow the exudative phase and can be gently debrided if necessary. Other complications include dyspigmentation, alopecia, pseudoepitheliomatous hyperplasia, depressed scars, and tissue distortion (nail dystrophy or notching of cartilage).

Overall, cryotherapy is a slow treatment that requires multiple sessions over a few months. While the treatment does work, compliance with treatment is low because of pain. Generally, an interprofessional team of nurses and clinicians will produce the best results and provide consistent patient follow up. [Level 5]

Review Questions

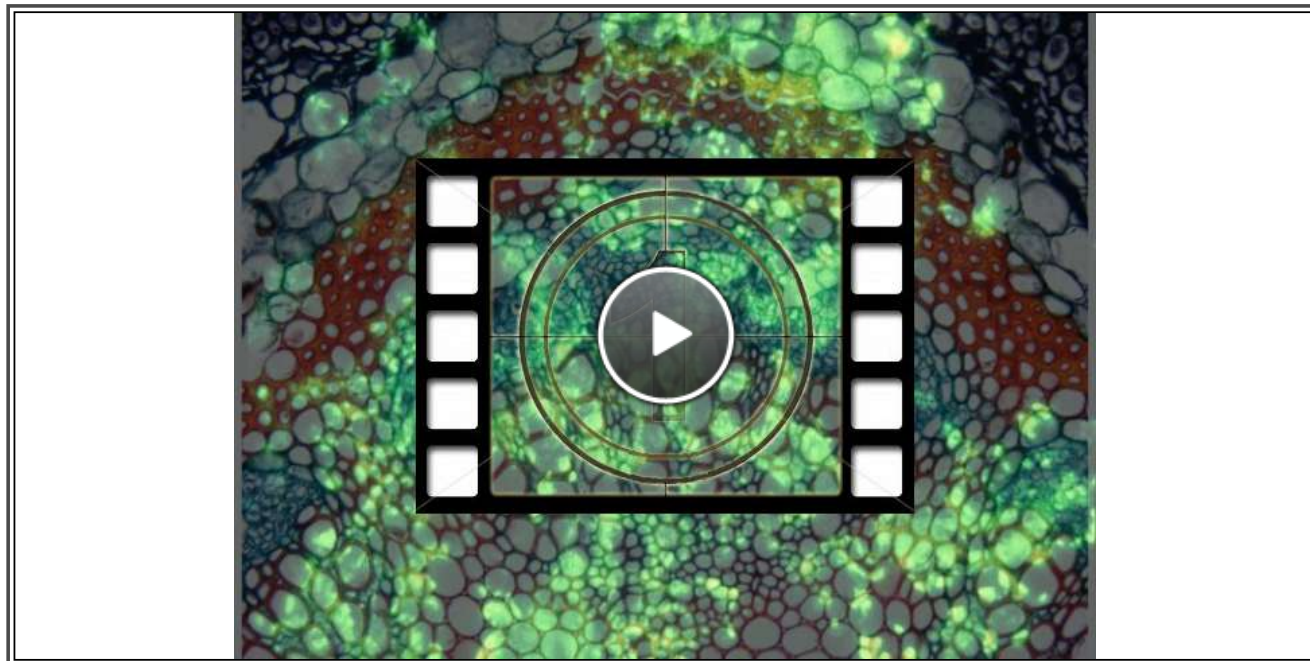
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Figures



[Download video file.](#) (838K, mp4)

Cryotherapy utilizing liquid nitrogen by cyogun. Contributed by Ahmad M. Al Aboud, M.D.

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CHAPTER 20

BIOMECHANICAL MEASUREMENTS

20.1. INTRODUCTION AND OBJECTIVES

In [Chapter 1](#) we covered the theoretical background to the mechanics of the human body. We saw how relatively simple mechanical models can be used to develop an insight into human performance. In this chapter we look at some of the problems involved in actually making *in vivo* measurements of forces and movements. The human body is a very complex structure and this, combined with the problems of gaining access to specific parts of the structure, can make it very difficult to make reproducible and meaningful measurements. This chapter should only be seen as the introduction to a very complex area of physiological measurement. These are some of the questions we will address:

- Can we describe human gait in terms of a few measurements?
- Is there such a thing as normal gait?
- Does standing cause ischaemia?
- What are normal static and dynamic loads during walking?

When you have finished this chapter, you should be aware of:

- How to measure joint angles.
- How a load cell works.
- How to measure limb positions during gait.
- How to measure foot pressure distribution.

It will be useful if you have read Chapter 1 before reading this chapter. It will also be helpful to have read the section on transducers in [Chapter 9](#) (section 9.5).

20.2. STATIC MEASUREMENTS

We will start by looking at the vertical forces involved in standing. A typical adult male might weigh (have a mass of) 70 kg and this will obviously give rise to a vertical force between the feet and the ground when standing. The force will be $9.81 \times 70 = 686.7$ N. A value of 9.81 has been assumed for g , the acceleration due to gravity. There is actually about a 0.6% difference in g between the equator and the poles so the force would be slightly less at the equator.

The force between the feet and the ground will give rise to a pressure or stress, which we can calculate if we know the area of contact between the feet and the ground. A typical area of contact for each foot is

0.015 m² so we can calculate the average pressure as:

$$\text{pressure between feet and ground} = 686.7/0.03 = 22\,890 \text{ N m}^{-2} = 22.9 \text{ kPa}.$$

This is the average pressure distributed over the surface of the feet. There will, of course, be spatial variations of the pressure as the contact between the feet and the ground is not uniform. The maximum pressure may well be a factor of five or more greater than the average pressure. We will return to the subject of the pressure distribution under the sole of the foot in section 20.2.2.

In order to set the average pressure between the feet and ground in context we can compare it with a normal arterial pressure of 16 kPa (120 mmHg) during cardiac systole and 10.6 kPa (80 mmHg) during diastole. The average pressure under the feet is higher than the arterial pressures so that we can expect standing to cause ischaemia in the tissues of the feet. This is indeed the case and explains why people move around when standing and change the weight distribution from foot to foot. This allows the blood to reach the tissues from time to time.

Many problems arise as a result of the pressure between the feet and the ground. However, many more problems arise in the joints between bones and in many cases these problems are caused by the high stresses involved. These stresses arise from the weight of the body itself, from the acceleration of body segments and from the musculature which holds a joint in place. In order to estimate the pressures which will arise in joints we need to know the areas of contact involved. Unfortunately, this is not easy to estimate and there are no satisfactory methods of measuring the stresses in joints *in vivo*. Brown and Shaw (1984) quote stress values measured *in vitro* for the acetabulum of the hip joint. The hip is a ball and socket joint; the ball is the head of the femur and the socket is a region on the edge of the hipbone called the acetabulum ('vinegar cup'). They quote maximum forces of 2100–2700 N and stresses of 5.3 MPa corresponding to an area of about 5 cm². The stresses are very high and explain the important role of the articular surfaces and the surrounding soft tissues, which in normal circumstances adapt and regenerate to carry the high stresses. In osteoarthritis the tissues are unable to adapt and regenerate as required so that the joint degenerates. Artificial joints have been developed to treat osteoarthritis and other pathological conditions of the joints. These joints have of course to be designed to withstand the same very high stresses as are encountered in a normal joint.

All of the forces we have considered so far are the static forces involved as a result of body weight. In section 20.3 we will consider the additional forces which arise when we accelerate parts of the body.

20.2.1. Load cells

Bathroom scales measure the vertical force exerted by the body when standing. They either contain a direct mechanical action involving a spring, or a transducer which converts force into an electrical signal which can be displayed digitally. The transducer can be referred to as a *load cell* and it contains a *strain gauge*. Load cells can be incorporated into a *force plate* and used to measure the forces involved in both standing and walking (see [figure 20.1](#)). By using three load cells as shown in [figure 20.1](#) it is possible to measure the lateral as well as the vertical forces involved when a person walks across the force plate. Force plates or platforms typically have sensors at the four corners of a rectangular plate. If each of the sensors contains transducers to measure force in three orthogonal directions, then it is possible to determine the vertical and two shear forces of stress applied to the surface of the platform. It is also possible to determine the point of application of the force and the moment around the axis perpendicular to the surface of the plate.

The lower part of [figure 20.1](#) shows some typical forces during normal walking. These are expressed as a fractional percentage of body weight and show that the vertical force can exceed body weight during heel strike and kick-off. During normal walking the peak vertical forces are up to 1.5 times body weight but during athletic events such as high jumping they can be 10 times body weight.

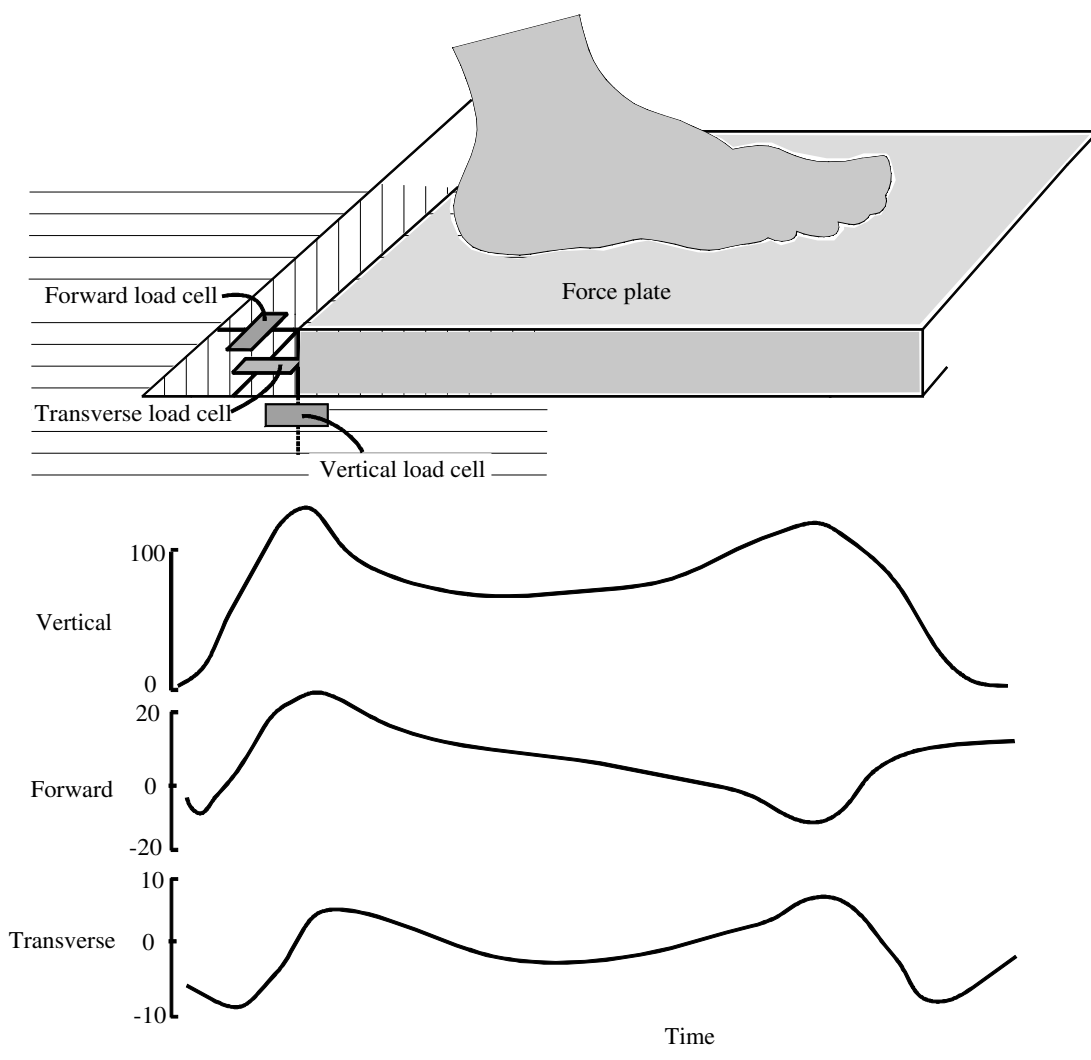


Figure 20.1. A force plate able to measure vertical, transverse and forward forces during walking. Load cells are shown at only one corner of the plate, but in practice they would usually be placed at all four corners. The lower part of the figure shows typical values for the three components of force obtained during walking. The results are expressed as a percentage of body weight.

20.2.2. Strain gauges

Strain gauges are a type of transducer (see section 9.5). They are almost always used to convert a mechanical change into an electrical signal. The range of strain gauges is very wide and indeed you may often not think that a particular technique is a strain gauge. For example there are now many laser ranging systems that can be used for measuring biological deformations. They form a type of strain gauge. We will consider just two types of very commonly used gauges: piezoelectric and foil gauges.

Piezoelectric strain gauges

The load cell attached to a force plate contains a strain gauge. Now, of course, we actually want to measure the stress on the plate so the input to the load cell is a stress and the output is an electrical signal proportional to the strain produced within the load cell.

Load cells commonly use a piezoelectric element as the sensor. The expression piezoelectric derives from the Greek *piezein* meaning to press. It was discovered in the 1880s that quartz and some other crystals produce electrical signals in response to mechanical pressure and much more recently various ceramic materials have been developed which exhibit the same effect, but to a much more marked degree. Lead titanate zirconate is a widely used piezoceramic. We can describe the properties of a piezoelectric material as follows.

The mechanical strain s that an elastic material experiences when subject to a stress σ is, according to Hooke's law, given by

$$\sigma = Ys \quad (20.1)$$

where Y is Young's modulus. If an isotropic dielectric material with permittivity ϵ is placed in an electric field E then a dielectric displacement D occurs, which is given by

$$D = \epsilon E. \quad (20.2)$$

However, piezoelectric materials are anisotropic and their permittivity and Young's modulus depend upon the direction and magnitude of the stress σ . This interaction can be shown to be described by the following four equations:

$$D = p\sigma \quad (20.3)$$

$$s = pE \quad (20.4)$$

$$s = kD \quad (20.5)$$

$$E = -k\sigma \quad (20.6)$$

where p and k are constants for the piezoelectric material.

From equations (20.3) and (20.4) we can see that the constant p can be seen either as the quotient of the electrical charge (at constant field) produced by a mechanical stress σ , or as the quotient of the strain s produced by an electric field. We can see that these equations describe the fact that piezoelectric materials produce a potential when subject to stress but will also do the reverse and produce a strain when a potential is applied to them. The units of p are either C N^{-1} or m V^{-1} .

From equations (20.5) and (20.6) we can see that the constant k can be seen either as the quotient of the electric field developed when stress σ is applied, or as the quotient of the strain s produced by the application of a dielectric displacement. The constants p and k are important parameters which describe the performance of a piezoelectric transducer. High values are usually best as they correspond to a high sensitivity.

Piezoelectric strain gauges can be used either to produce an output charge in proportion to an applied force as described by equation (20.3), or to produce an output voltage as described by equation (20.6). The former method is normally used because the sensitivity of a voltage amplifier is affected by the capacitance of the input cables and the low-frequency response of the system is limited by leakage of the charge through the piezoelectric material. The type of circuit usually employed is illustrated in [figure 20.2](#), which shows a charge amplifier configuration.

Using the terminology illustrated in figure 20.2 we can use Kirchhoff's law to sum the currents at the inverting input of the operational amplifier to give:

$$i = i_t + i_m - i_f$$

therefore

$$\int i \, dt = \int (i_t + i_m - i_f) \, dt.$$

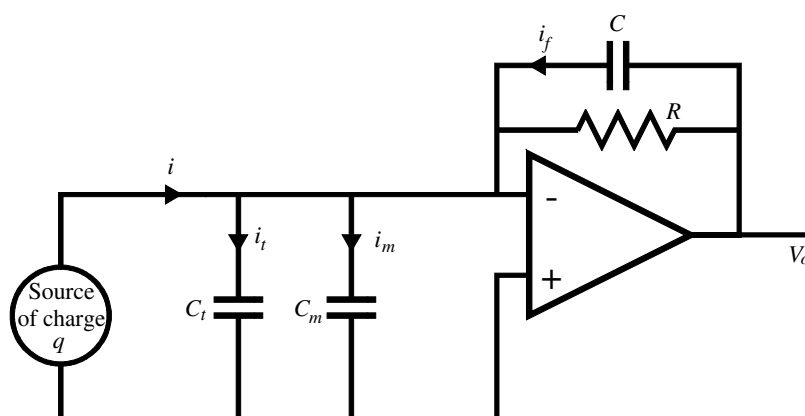


Figure 20.2. Circuit diagram of a charge amplifier. C_t is the capacitance of the transducer and C_m that of the measurement system.

Now as the operational amplifier has infinite gain there is no significant input voltage and hence no potential can accumulate across C_t and C_m . Therefore,

$$\int (i_t + i_m) dt = 0$$

and

$$V_o = \frac{\int i dt}{C} = -\frac{q}{C}. \quad (20.7)$$

The output is thus proportional to the accumulated input charge q and the feedback capacitance C and independent of C_t and C_m . We have ignored the resistance R in this calculation.

The circuit shown works very well, but when used for measuring slowly changing forces drifts can become very significant. It is left to the reader to calculate the drift which will arise from the bias current flowing into the inverting input of the operational amplifier. This current will be integrated by the feedback capacitance to give a steady drift in the output voltage. This drift can be reduced by including a feedback resistance R as shown, but this will reduce the low-frequency response of the circuit. Transient charges, such as those which will occur when a subject walks over a force plate, will be recorded but very slowly changing forces will not appear.

Foil strain gauges

It was noticed by Lord Kelvin in about 1850 that the resistance of a metal wire increases with increasing strain and that different materials have different sensitivities to strain. What was being observed is simply that the resistance of a given volume of metal will depend upon its shape. If we consider a wire of circular cross-section area a as shown in figure 20.3(a) then its resistance (R) is given by $\rho l/a$ and its volume by al . However, if we keep the volume of the wire constant but double the length to $2l$ (figure 20.3(b)) then the resistance will become $(\rho 2l)/(a/2) = 4R$. The resistance has quadrupled as the length had doubled. If we use the same volume of conductor to form two sections of wire with cross-sections $a/2$ and $3a/2$ (20.3(c)) but each of length $l/2$, then the total resistance of the two sections will be $1.33R$.

Electrical resistance or foil strain gauges are made by laying down the metallic conductor onto a backing material. The sensitivity of the gauge depends upon the configuration of the conductor and the material used

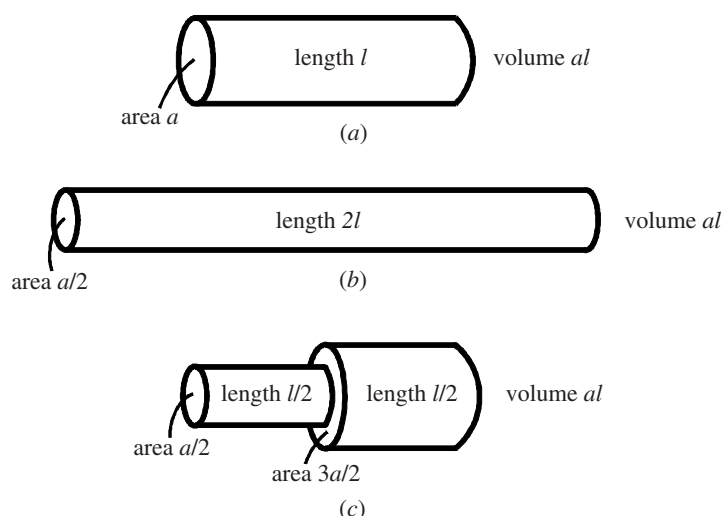


Figure 20.3. Three sections of wire all with the same volume.

for the backing. Sensitivity is defined in terms of a gauge factor (GF) which is given by

$$GF = (\Delta R/R)/\text{strain}. \quad (20.8)$$

The gauge factor in the example we gave above was 2 and this is the maximum which is usually obtained.

Electrical resistance strain gauges are affected by temperature simply because metals have a resistivity which increases with increasing temperature. The temperature coefficient of copper is about $0.5\% \text{ } ^\circ\text{C}^{-1}$. This is a very high temperature coefficient and makes copper unsuitable for use in strain gauges. However, various alloys are available with very much lower temperature coefficients. Another way to avoid temperature effects is to place two conductors on the foil in positions where the resistance will increase with strain in one case but decrease in the other. This can often be achieved by placing the conductors on opposite sides of the backing material. If the two resistances are then placed in opposite arms of a bridge configuration then changes with temperature will tend to cancel, whereas the changes with strain will sum together. The principle is the same as that used to provide temperature compensation in pressure transducers (see section 18.4).

20.2.3. Pedobarograph

Measurement of the pressure distribution under the feet is a vital component of the assessment of standing and gait. One way in which to make the measurements is to construct a two-dimensional matrix of force transducers on which the subject stands. This technique certainly works, but it is relatively expensive and complex because a large number of transducers are required. If a spatial resolution of 2 mm is required over a force plate area of 30×30 mm then 225 transducers are required and the associated electronics to interrogate them. An elegant solution to this problem was suggested by Chodera in 1957.

Chodera suggested the system illustrated in [figure 20.4](#). A thick glass plate is edge illuminated as shown such that the light will be totally internally reflected within the glass. Total internal reflection takes place because the refractive index of the glass is higher than that of the surrounding air (see Snell's law in section 3.5.1). On top of the glass plate is a sheet of plastic material whose refractive index is higher than that of the glass. If this sheet of material makes optical contact with the glass then the total internal reflection

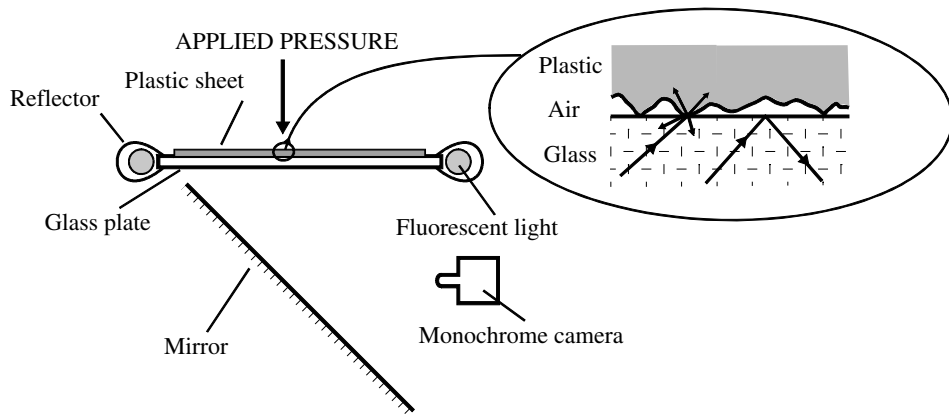


Figure 20.4. *The principle of a pedobarograph which uses an edge-illuminated glass sheet and a covering plastic sheet to measure pressure distribution.*

will be destroyed and hence light will escape and be scattered by the surface of the sheet of material. If the plate of glass is viewed from the underside then the light scattered by the plastic sheet will be seen. Now if the surface texture of the plastic sheet is such that increasing pressure will give a proportional increase in the area of optical contact with the glass then the intensity of light seen from underneath will be proportional to pressure. The performance of such a system depends very much on the plastic sheet on which the subject places their feet. The material must deform in a uniform fashion and must behave in a reversible fashion when pressure is released.

Calculation of how the area of contact between the glass plate and the plastic sheet might change with applied pressure is not a trivial problem. It was shown in [Chapter 1](#) that many materials exhibit a stress/strain relationship as shown in figure 20.5. The relationship is linear until the yield point Y is reached, but beyond this point the slope of the curve drops off very quickly.

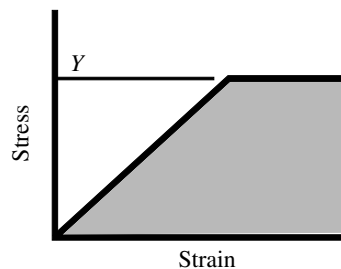


Figure 20.5. *General form of the stress versus strain relationship.*

If the surface of the material is considered to be made up of pyramids as shown in [figure 20.6](#) then we can consider what might happen when pressure P is applied over unit area. A force P will have to be supported by the pyramid of deformable plastic material.

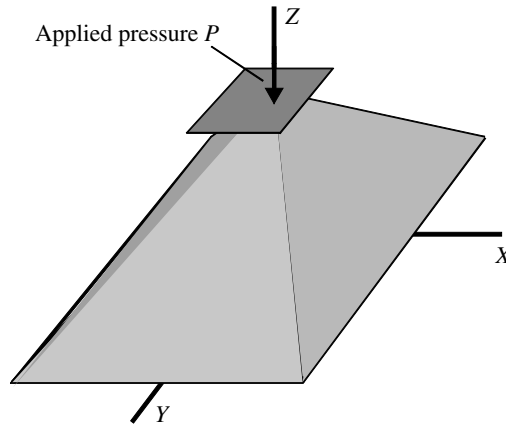


Figure 20.6. Pressure P applied to a pyramid of deformable material.

Now the apex of the pyramid will have zero area and hence the applied stress will be infinite. The yield point of the material will be exceeded and so the strain will increase rapidly such that the apex of the pyramid will collapse. It will collapse to the point where the stress no longer exceeds the yield point Y . This point will be given by

$$P = Y \times \text{area of contact.}$$

We can therefore expect the area of contact to be proportional to the applied pressure. This is what is found to be approximately true in practice. However, a more detailed investigation (see Betts *et al* 1980) shows that as pressure increases the number of points of contact between the plastic sheet and the glass increases. This is not unreasonable because the surface of the plastic sheet will be irregular with protrusions above the surface showing a distribution of heights. One of the problems with this technique is that the response time is quite slow and hysteresis also occurs. However, Betts *et al* state that by careful selection of the plastic material a response to an increasing applied pressure occurs within less than 5 ms and to a decreasing pressure within about 20 ms. The longer response time to a decreasing pressure is to be expected if the material is being deformed beyond its yield point.

Figure 20.7 shows measurements of mean pressure over seven regions of interest under a normal foot during walking. It is easy to identify the times of heel-strike and toe-off. It can also be seen that the pressure distribution is very non-uniform and changes during gait. The type of measurement shown has been widely used to investigate gait and to observe the effects of abnormal gait following surgery to the lower limbs.

20.3. DYNAMIC MEASUREMENTS

We now turn from static forces to the forces that result from accelerations of the body. Our musculoskeletal system maintains body posture, but of course posture is a function of time t . We can represent the position vector x of a part of the body and then say that the forces $F(t)$ as a function of time are given by:

$$a_2(x) \ddot{x} + a_1(x) \dot{x} + a_0(x)x = F(t)$$

where a_2 represents inertial forces, a_1 represents energy losses such as viscous losses and a_0 represents stiffness. In reality a_0 , a_1 and a_2 will be functions of x so that the system is nonlinear. For small changes we can often assume linearity and if we are to understand the system then we need to be able to measure a_0 , a_1 and a_2 . We gave stress values in section 20.1 which showed that the static forces are often exceeded by the dynamic ones, but to quantify these we must be able to measure displacements, velocities and accelerations.

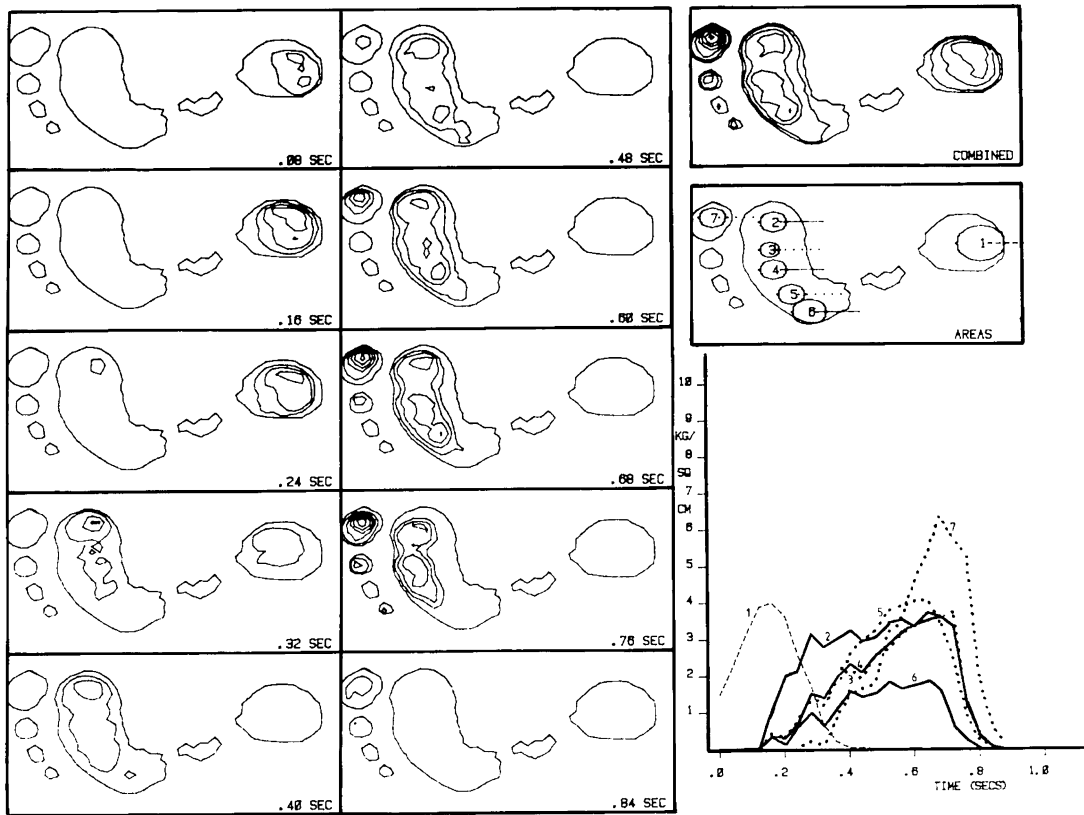


Figure 20.7. A typical dynamic isopressure contour plot output obtained when a normal subject walks across the plate of a pedobarograph. On the left are shown 10 selected frames equally distributed from throughout the footstep sequence. At the top right is the combined frames image followed by the position of the seven selected areas of interest, and finally the pressure/time curves ($1 \text{ kg cm}^2 = 98.1 \text{ kPa}$). (Image kindly provided by Dr R P Betts, Royal Hallamshire Hospital, Sheffield.)

20.3.1. Measurement of velocity and acceleration

It can be argued that it is necessary to measure only one of the three variables, displacement, velocity and acceleration, because the others can be found by differentiation or integration. However, integration has the problem that constants of integration are unknown and differentiation can produce a large amount of high-frequency noise. It is best to measure the variable required as directly as possible.

Relative velocity (v) can be measured by moving a coil through a steady magnetic field (H) which will give rise to an induced voltage u where $u \propto H v$. The problem with this approach is in producing a steady field H over the volume in which the movement takes place. The technique can be used to measure relative velocity between two limb segments by attaching a permanent magnet to one segment and a coil to the other. However, the output is very nonlinear as the field is not constant around the permanent magnet. If absolute velocity is required then it is often best to integrate the output from an accelerometer.

Accelerometers usually consist of a small mass m attached to a force transducer as shown in [figure 20.8](#). Acceleration a will give rise to a force am on the piezoelectric transducer. It is worth noting that the

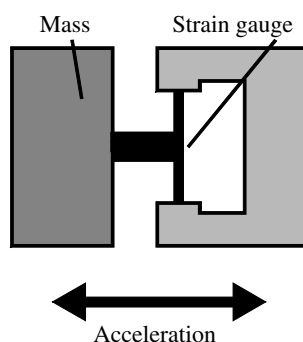


Figure 20.8. *Basis of an accelerometer. Acceleration in the direction shown will give rise to a proportional force on the piezoelectric strain gauge transducer.*

accelerometer cannot distinguish between the force mg due to gravity and the force caused by acceleration of the mass m . If the accelerometer is rotated within the Earth's gravitational field then an output will be obtained. This can be a very significant source of error if such a device is used to measure the acceleration of parts of the body.

20.3.2. Gait

People are able to move around with relative ease, but exactly how do we walk? How do we put one foot in front of the other and why does it become very difficult for some people? The science of human walking is often called gait analysis. The basic mechanics of walking and running were described in section 1.7.2 of [Chapter 1](#). However, in order to study anything it is necessary to be able to make measurements and in gait analysis these measurements are quite difficult. They fall into three main categories: techniques to measure the forces exerted on the ground during walking, instruments to measure muscle activity and instruments to measure the relative positions of the limbs and the movement of the body in space. We have already described some of the techniques for measuring the forces between the feet and the ground and some of the techniques for measuring muscle activity were covered in [Chapter 16](#). We will now give some thought as to how can we describe body position and then how we can measure these descriptors. [Table 20.1](#) gives some of the factors which have been used to describe normal walking. These parameters need careful definition if comparisons are to be made between measurements made in different laboratories. The stance and swing phases are defined for one foot and correspond to the time when the foot is either in contact with the ground or off the ground, respectively.

It is a struggle to separate the males and females using the data given in [table 20.1](#). In the last column of the table the difference between the mean measurement for males and females has been divided by the mean standard deviation on the measurements. This gives an indication of the overlap between the distributions. A value of 2 would mean that there was an overlap of about 15% if the distributions were normally distributed. Step length and hip flexion would appear to be the best discriminants between male and female gait—not a surprising conclusion.

A considerable amount of the variation in the measurements given in [table 20.1](#) may be accounted for by variations with age. Older people will certainly walk more slowly than younger ones. The variation with age is well illustrated by the data in [table 20.2](#), which gives the variations in six different gait parameters over the first 7 years of life.